Nonparametric Drift Estimation for Stochastic Differential Equations

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Abstract

We consider N i.i.d. time-homogeneous diffusion processes $(X_t^i, i = 1, \dots, N)$ with drift $b(x)$ and diffusion coefficient $\sigma(x)$ such that $\sigma(x)$ is known from observations of the sample paths. Our concern is the nonparametric estimation of the drift function b from continuous observations of the sample paths (X_t^i) through a fixed time interval $[0, T]$. We derive some regularity assumptions via some Kusuoka-Stroock bounds on the density of the SDE before providing a Nikol'skii condition of the averaged density for the risk-bounds. We construct a Nadaraya-Watson estimator of the drift function and provide the L^2-risk bounds on this nonparametric estimator. We discuss some data-driven strategies for bandwidth selection via a Leave-one-out Cross validation method (LooCV) and a Penalized Comparison to Overfitting method (PCO). We also provide some insight into some model extensions for space-time dependent diffusions.

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Statement of Originality

Although this essay is largely based on a review of the existing literature, unless explicitly stated otherwise, all remarks, examples, and explanations presented are my own.



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Notations

- We set $a \wedge b := \min\{a, b\}$ and $a \vee b := \max\{a, b\}$.
- We set $x_+ := \max\{0, x\}$ to be the positive part of x.
- For every $p \geq 1$, the space $L^p(\mathbb{R}, dx)$ is equipped with the norm $\|\cdot\|_p$ where

$$\|\varphi\|_p := \left(\int_{\mathbb{R}} \varphi(x)^p dx\right)^{\frac{1}{p}}; \qquad \forall \varphi \in L^p(\mathbb{R}, dx).$$

- For a given kernel δ, the norm $\|\cdot\|_{2,\delta}$ on the space $L^2(\mathbb{R}, \delta(x)dx)$ is given by

$$\|\varphi\|_{2,\delta} := \left(\int_{\mathbb{R}} \varphi(x)^2 \delta(x) dx\right)^{\frac{1}{2}}; \qquad \forall \varphi \in L^2(\mathbb{R}, \delta(x)dx).$$

- For every $\varphi \in L^2(\mathbb{R}, \delta(x)dx)$ and $A, B \in \mathbb{R}$ such that $A < B$, we define the following truncated norm

$$\|\varphi\|_{\delta,A,B} := \|\varphi \mathbb{1}_{[A,B]}\|_{2,\delta} \equiv \left(\int_A^B \varphi(x)^2 \delta(x) dx\right)^{\frac{1}{2}}; \qquad \forall \varphi \in L^2(\mathbb{R}, \delta(x)dx).$$

- We set $C_0(\mathbb{R})$ to be the space of all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that $\lim_{|x|\to\infty} f(x) = 0$ and equipped with the uniform norm

$$\|\varphi\|_\infty := \sup_{x\in\mathbb{R}} |\varphi(x)|; \qquad \forall \varphi \in C_0(\mathbb{R}).$$

- $\mathbb{H}^2$ is the space of stochastic processes $(Y_t)_{t\in[0,T]}$ adapted to the filtration generated by the Brownian motion $W := (W_t)_{t\in[0,T]}$, and such that

$$\int_0^T \mathbb{E}\big(Y_t^2\big) dt < \infty.$$

Here, a process $(Y_t)_{t\in[0,T]}$ defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ is said to be **adapted** to the filtration $(\mathcal{F}_t)_{t\in[0,T]}$ (i.e. a collection of sub-sigma-algebras of $\mathcal{F}$ such that $\mathcal{F}_s \subseteq \mathcal{F}_t$ for all $s \leq t$) if Y_t is an $\mathcal{F}_t$-measurable random variable at each time $t \in [0, T]$. Intuitively, this just guarantees that the value of the process Y_t is observable at time t.

1 | Introduction

Stochastic Differential Equations (SDEs) are a natural choice to model the time evolution of dynamical systems which are subject to random influences. For instance, in physics the dynamics of ions in supersonic conductors are modelled via Langevin equations, or in engineering the dynamics of mechanical devices are described by differential equations under the influence of process noise as error measurements [1]. SDEs are fully characterized by two components, the drift function b that can be understood as the (infinitesimal) conditional mean of the process and the diffusion σ corresponding to the (infinitesimal) conditional variance [2, pp.5]. Consequently, the construction of estimators for the drift and diffusion coefficients is a widely investigated inference problem that has been studied both in parametric and nonparametric frameworks.

Since the 1980's, much of the early theory concerned the parametric estimation of SDEs for which the Itô SDE is of the form $dX_t = b(t, X_t, \theta)dt + \sigma(t, X_t)dW_t$, $t \geq 0$, for some unknown parameter θ and a Brownian motion W_t. The parametric setup for SDEs is largely developed within an asymptotic framework, where inference is studied under specific regimes such as increasing observation horizons or increasingly fine sampling of the process. In these settings, one seeks to construct parameter estimators that are both theoretically well behaved and practically meaningful. Approaches via likelihood-based and Bayesian methods serve as natural tools for this analysis and allow for a rigorous study of key statistical properties, including consistency and asymptotic normality (see S.Särkkä & A.Solin [3]). In practice, however, likelihood-based methods are often limited by the lack of closed-form transition densities since the associated Fokker–Planck equation is rarely solvable, especially in nonlinear diffusion models. Consequently, inference typically relies on approximate discretization schemes, such as the Euler–Maruyama or Milstein methods, whose theoretical properties must be carefully assessed (see [3]). On the other hand, nonparametric estimation provides a more flexible alternative by allowing the drift and diffusion functions to be estimated without specifying a particular functional form where we now consider a SDE $dX_t = b(t, X_t)dt + \sigma(t, X_t)dW_t$, $t \geq 0$. In this framework and until the end of the 1990's, all the estimators for the drift function were computed from a single long-time observation of the ergodic stationary solution of the SDE (see Y.Kutoyants [4]) [5]. Recently, attention has shifted to drift estimators based on multiple observations of the diffusion process allowing for a Functional Data Analysis perspective. In this setting, the proposed estimators are typically constructed using tools from regression, such as projection-based or kernel methods.

This work is divided into three parts. In Chapter 2, we briefly present the construction of general local polynomial estimators of arbitrary order k before formally defining the Nadaraya-Watson (NW) estimator as an estimator of order $k = 0$. We end the chapter by a crucial remark on a slightly different vision on the construction of the NW estimator that will be useful to understand the construction of the nonparametric drift estimator. In Chapter 3, we present some regularity conditions on the density function $p_t(x_0, x)$ of the SDE via some Kusuoka-Stroock bounds. We establish these conditions in order to present a Nikol'skii type bound on a density function that will be crucial to derive the risk-bounds of the NW estimator. Finally, in Chapter 4, we construct the NW estimator of the drift function and derive the corresponding risk-bounds. We then end the chapter by briefly presenting how one would perform bandwidth selection via the Leave-one-out Cross Validation (LooCV) and Penalized Comparison to Overfitting method (PCO). We also mention some model extensions where we extend the framework of time-homogeneous diffusions to space-time dependent SDEs.

2 | Local Polynomial Regression & the Nadaraya-Watson Estimator

In this chapter, we aim to briefly present the construction of the well-known Nadaraya-Watson nonparametric regression estimator. As a first step, we recall the problem of nonparametric regression via local polynomial regression before presenting this general procedure for the estimation of the particular function $f(x) = \mathbb{E}(Y|X = x)$, where (X,Y) is a pair of random variables and $x \in \mathbb{R}$. The results and constructions in this chapter have been widely investigated in the nonparametric estimation literature, and we will closely follow the work of Alexandre B. Tsybakov [6] and of Larry Wasserman [7].

The goal is to find an estimator for the *regression function* $f : \mathbb{R} \to \mathbb{R}$ *of* Y *on* X (assumed to be smooth) given by $f(x) = \mathbb{E}(Y|X = x)$ such that (X,Y) is a pair of real-valued random variables with the condition $\mathbb{E}(|Y|) < \infty$. To this end, we will estimate the function f using a sample of n i.i.d. pairs of random variables $\{(X_i,Y_i)\}_{i=1}^n$ following a common distribution given by the one of (X,Y). Since we are considering a nonparametric framework, we have that f is an element of a nonparametric class of functions. In our context, we also remark from the way the regression function has been designed, we may consider a model with a conditional residual $\xi := Y - \mathbb{E}(Y|X)$. This quantity is being used to assess how well a model fits the data by the estimation of this random error. Thus, the observations may be seen as *noisy* by considering the model $Y_i = f(X_i) + \xi_i$ for $i \in \{1,...,n\}$, where $\xi_i \overset{\text{iid}}{\sim} \xi$ are centred errors.

In a general least-squares method, the task is to minimize the sum of squared residuals $\sum_{i=1}^n \xi_i^2$. However in the setup of our noisy model, we will instead minimize a weighted least-squares in order to apply a *local polynomial regression*. The intuition is the following: aiming to estimate the regression function $f(x) = \mathbb{E}(Y|X = x)$ at every point $x \in \mathbb{R}$, from the smoothness of f we have that the most informative data points X_i will be those being sufficiently close to x. As a result, it will make sense to use a kernel K having the form $K\big(\frac{X_i - x}{h}\big)$, for some bandwidth $h > 0$, as the corresponding weights for the regression. Here, a **kernel** is a symmetric, smooth function $K(x)$ such that for all $x \in \mathbb{R}$, $K(x) \geq 0$, $\int_{\mathbb{R}} K(x)dx = 1$, $\int_{\mathbb{R}} xK(x)dx = 0$ and $\int_{\mathbb{R}} x^2K(x)dx > 0$. For example, a well-known kernel used in nonparametric regression is $K(x) = (1 - |x|^3)^3 \mathbb{1}_{\{|x|<1\}}$ (see the *tricube kernel* in [8]) which is a peaked function at $x = 0$ with vanishing tails for $|x| > 1$, thus favouring the X_i being in some relatively small neighbourhood of x. Now that we have formally given the structure of the kernel, it remains to appropriately choose the form of the residual terms. For this, we first consider the following class of functions:

Definition 2.0.1. *Let T be an interval in $\mathbb{R}$ and set β, L to be two positive numbers. We define the* ***Hölder class*** *$\Sigma(\beta, L)$ on T to be the set of all $k = \lfloor \beta \rfloor$ times differentiable functions $f : T \to \mathbb{R}$ whose derivative $f^{(k)}$ satisfies,*

$$\left| f^{(k)}(x) - f^{(k)}(y) \right| \leq L|x - y|^{\beta - k}, \qquad \forall x, y \in T. \tag{2.1}$$

Now, consider the function $f \in \Sigma(\beta, L)$ defined previously with $\beta > 1$ and $k = \lfloor \beta \rfloor$. Thus we know that f is k-times differentiable, and by Taylor's Theorem there exists a k-th order Taylor polynomial of f such that, for $z \to x$:

$$f(z) = \sum_{i=0}^{k} \frac{f^{(i)}(x)}{i!}(z-x)^i + o\big(|z-x|^k\big) = \theta^T(x) U\left(\frac{z-x}{h}\right) + o\big(|z-x|^k\big),$$

where $U(u) = \left(1, u, \frac{u^2}{2!}, ..., \frac{u^k}{k!}\right)^T$ and $\theta(x) = \left(f(x), f'(x)h, f''(x)h^2, ..., f^{(k)}(x)h^k\right)^T$. As a result, by taking only the k-th order Taylor polynomial of f as the approximation of f we may formally give the definition of a local polynomial estimator of $f(x)$:

Definition 2.0.2. *Let $K : \mathbb{R} \to \mathbb{R}$ be a kernel, $h > 0$ be a bandwidth, and $k \geq 0$ be an integer. A **local polynomial estimator of order k of $\theta(x)$** (or **LP(k) estimator of $\theta(x)$** for short) is a vector $\hat{\theta}_n(x) \in \mathbb{R}^{k+1}$ defined by:*

$$\hat{\theta}_n(x) = \underset{\theta \in \mathbb{R}^{k+1}}{\arg\min} \sum_{i=1}^{n} \left[Y_i - \theta^T(x) U\left(\frac{X_i - x}{h}\right)\right]^2 K\left(\frac{X_i - x}{h}\right). \tag{2.2}$$

*The corresponding **local polynomial estimator of order k of $f(x)$** (or **LP(k) estimator of $f(x)$** for short) is given by the statistic,*

$$\hat{f}_n(x) = U^T(0)\hat{\theta}_n(x). \tag{2.3}$$

Note that the $LP(k)$ estimator of $f(x)$ is simply the first coordinate of the vector $\hat{\theta}_n$. Now, a very important property of local polynomial estimators is that they are linear estimators, i.e. we can write $\hat{f}_n(x) = \sum_{i=1}^{n} Y_i \widetilde{W}_{ni}(x)$ for some appropriate weights $\widetilde{W}_{ni}(x)$ (a proof of this fact is shown in Appendix (B.1)). We are now ready to provide the definition of the Nadaraya-Watson estimator, since it is the special case of the $LP(0)$ estimator. It is in fact a local constant least-squares approximation of the outputs Y_i.

Definition 2.0.3. *Let $K : \mathbb{R} \to \mathbb{R}$ be a kernel and $h > 0$ be a bandwidth. The **Nadaraya-Watson estimator** is the $LP(0)$ estimator. Thus, it is defined by:*

$$f_n^{NW}(x) = \underset{\theta \in \mathbb{R}}{\arg\min} \sum_{i=1}^{n} (Y_i - \theta)^2 K\left(\frac{X_i - x}{h}\right). \tag{2.4}$$

We may also give the explicit formula for the Nadaraya-Watson estimator by directly solving the minimisation problem (a proof of this fact is shown in Appendix (B.2)).

Proposition 2.0.4. *Let $K : \mathbb{R} \to \mathbb{R}$ be a kernel and $h > 0$ be a bandwidth. The Nadaraya-Watson estimator is given by:*

$$f_n^{NW}(x) = \frac{\sum_{i=1}^{n} Y_i K\left(\frac{X_i - x}{h}\right)}{\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right)} \cdot \mathbb{1}_{\left\{\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right) \neq 0\right\}}. \tag{2.5}$$

As expected, we note that the Nadaraya-Watson estimator is an example of a *linear nonparametric regression estimator* since it can be interpreted as a weighted sum of the Y_i's. Indeed:

$$f_n^{NW}(x) = \sum_{i=1}^{n} Y_i \left(\frac{K\left(\frac{X_i - x}{h}\right)}{\sum_{j=1}^{n} K\left(\frac{X_j - x}{h}\right)} \cdot \mathbb{1}_{\left\{\sum_{j=1}^{n} K\left(\frac{X_j - x}{h}\right) \neq 0\right\}} \right) \equiv \sum_{i=1}^{n} Y_i W_{ni}^{NW}(x),$$

which is such that the weights satisfy $\sum_{i=1}^{n} W_{ni}^{NW}(x) = 1$. We remark that these weights are (Y_i)-independent

and that they are fully determined by the constants n, i, x and also by the observed values of the random vector $\mathbf{X} = (X_1, ..., X_n)$.

Remark 2.0.5. *Let us end this chapter by the following crucial remark based on [9]. In order to construct the Nadaraya-Watson estimator, we have chosen to present it as a special case of a local polynomial estimator. However, for the purpose of the later sections, we may view this construction from another angle which has been initiated by* ***Rosenblatt*** *(1956). Aiming to find an estimator for a density $f(x)$, consider the empirical estimator $\hat{F}(x) := \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{\{X_i \leq x\}}$ of the cumulative distribution function $F(x)$ for a given sample $\{X_i\}_{i=1}^n$. Then, since for all $h > 0$*

$$F(x+h) - F(x-h) = \mathbb{P}(x-h < X_1 \leq x+h) = \int_{x-h}^{x+h} f(y)dy,$$

by taking h small enough we should expect $f(x) \approx \big(F(x+h) - F(x-h)\big)/(2h)$. We can then propose the following estimator of the density:

$$\begin{aligned}\hat{f}_h(x) := \frac{\hat{F}_n(x+h) - \hat{F}_n(x-h)}{2h} = \frac{1}{2nh}\sum_{i=1}^{n}\big(\mathbb{1}_{\{X_i \leq x+h\}} - \mathbb{1}_{\{X_i \leq x-h\}}\big) &= \frac{1}{2nh}\sum_{i=1}^{n}\mathbb{1}_{\{x-h < X_i \leq x+h\}} \\ &= \frac{1}{nh}\sum_{i=1}^{n}\frac{1}{2}\mathbb{1}_{\left\{-1 < \frac{X_i - x}{h} \leq 1\right\}} \\ &=: \frac{1}{nh}\sum_{i=1}^{n} K_0\left(\frac{X_i - x}{h}\right),\end{aligned}$$

where $K_0(u) := \frac{1}{2}\mathbb{1}_{\{-1<u\leq 1\}}$ is the rectangular kernel. A generalization has then been provided by ***Parzen*** *(1962) considering the density estimators*

$$\hat{f}_h^P(x) := \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right) \quad \text{and} \quad \hat{f}_h^P(x,y) := \frac{1}{nh^2}\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right) K\left(\frac{Y_i - y}{h}\right), \tag{2.6}$$

for some bandwidth $h > 0$ and kernel $K : \mathbb{R} \to \mathbb{R}$. How can this construction help us to recover the Nadaraya-Watson estimator? Aiming to estimate the regression function $g(x) = \mathbb{E}(Y|X = x)$, by definition of the conditional expectation and considering a kernel of order 1 (i.e. for $j \in \{0,1\}$, $u \mapsto u^j K(u)$ is integrable satisfying $\int_{\mathbb{R}} K(u)du = 1$ and $\int_{\mathbb{R}} uK(u)du = 0$), we find:

$$\hat{g}(x) = \frac{\int_{\mathbb{R}} y\hat{f}_h^P(x,y)dy}{\int_{\mathbb{R}} \hat{f}_h^P(x,y)dy} = \frac{\int_{\mathbb{R}} y\hat{f}_h^P(x,y)dy}{\hat{f}_h^P(x)} = \frac{1}{\hat{f}_h^P(x)}\left(\frac{1}{nh^2}\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right)\int_{\mathbb{R}} yK\left(\frac{Y_i - y}{h}\right)dy\right), \tag{2.7}$$

provided that $\hat{f}_h^P(x) \neq 0$. We now note that the final integral can be expanded in the following way:

$$\frac{1}{h}\int_{\mathbb{R}} yK\left(\frac{Y_i - y}{h}\right)dy = \int_{\mathbb{R}}\left(\frac{y - Y_i}{h}\right)K\left(\frac{Y_i - y}{h}\right)dy + \frac{Y_i}{h}\int_{\mathbb{R}} K\left(\frac{Y_i - y}{h}\right)dy.$$

Then, performing a change of variable we finally conclude

$$\frac{1}{h}\int_{\mathbb{R}} yK\left(\frac{Y_i - y}{h}\right)dy = -h\underbrace{\int_{\mathbb{R}} uK(u)du}_{=0} + Y_i\underbrace{\int_{\mathbb{R}} K(u)du}_{=1} = Y_i.$$

Thus, to estimate the regression function $g(x)$, we simply consider the Parzen-Rosenblatt estimator $\hat{f}_h^P(x)$ (respectively $\hat{f}_h^P(x,y)$) of the density (respectively joint density). Indeed, the estimator $\hat{g}(x)$ in equation (2.7) is exactly the one found in Proposition (2.0.4).

3 | Regularity Bounds & a Nikol'skii Condition for the Density

This chapter will be devoted to the presentation of the existence and regularity of the solutions to a Fokker-Planck equation associated to the stochastic differential equation $X_t = x_0 + \int_0^t b(X_s)ds + \int_0^t \sigma(X_s)dW_s$ for every $t \in (0,T]$, namely the density function $p_t(x_0,\cdot)$. In addition, we will be mostly interested in the function

$$f(x) := \frac{1}{T-t_0}\int_{t_0}^{T} p_s(x_0,x)ds$$

which will be the density of interest in the next chapter. The results presented below will be considered as assumptions for the rest of the essay, since they will be used to present a crucial Nikol'skii type condition on f. In order to do so, we will first present and motivate the Kusuoka-Stroock bounds on the density $(t,x) \mapsto p_t(x_0,x)$ and its space derivatives. We now state and prove a first Kusuoka-Stroock bound on the density p_t, where the general steps are from *Comte & Marie* [10, pp.4–5].

Proposition 3.0.1. *For any $t \in (0,T]$ and $x \in \mathbb{R}$, the density of X_t satisfies the following:*

$$p_t(x_0,x) \leq \frac{\Lambda_1}{t^{\frac{1}{2}}} \exp\left[-\Lambda_2 \frac{(x-x_0)^2}{t}\right], \tag{3.1}$$

for some positive constants Λ_1, Λ_2 dependent on T but independent of t and x.

Proof. For every $x \in \mathbb{R}$, consider the deterministic flow $t \mapsto \theta_t(x)$ to be the solution to the differential equation

$$y_t = x + \int_0^t b(y_s)ds, \qquad t \in [0,T]. \tag{3.2}$$

Assume also that σ satisfies a non-degeneracy condition where there exists $\alpha, A > 0$ such that $\alpha \leq |\sigma(x)| \leq \alpha + A$ for all $x \in \mathbb{R}$. By Menozzi et al. [11, Theorem 1.2], the transition density p_t is well-defined and there exists two constants $\overline{\Lambda}_1$ and $\overline{\Lambda}_2$, dependent on T but independent of x and t, such that for every $x, y \in \mathbb{R}$:

$$p_t(x,y) \leq \frac{\overline{\Lambda}_1}{t^{\frac{1}{2}}} \exp\left[-\overline{\Lambda}_2 \frac{(y-\theta_t(x))^2}{t}\right]. \tag{3.3}$$

Now, for any $t \in (0,T]$ and $x \in \mathbb{R}$, we observe the following:

$$\begin{aligned}
|x - x_0| &\leq |x - \theta_t(x_0)| + |\theta_t(x_0) - x_0| \\
&= |x - \theta_t(x_0)| + \left|\int_0^t b(\theta_s(x_0))ds\right| \qquad \rightsquigarrow \theta_t \text{ is solution of (3.2)} \\
&\leq |x - \theta_t(x_0)| + \int_0^t |b(\theta_s(x_0))|ds
\end{aligned}$$

$$\leq |x-\theta_t(x_0)| + t \sup_{s\in[0,T]} |b(\theta_s(x_0))|.$$

Thus by defining $S_T(x_0) := \sup_{s\in[0,T]}|b(\theta_s(x_0))|$ we conclude that $|x-x_0| \leq |x-\theta_t(x_0)| + tS_T(x_0)$. Furthermore, using the fact $(a+b)^2 \leq 2(a^2+b^2)$ we know that

$$|x-x_0|^2 \leq \big(|x-\theta_t(x_0)| + tS_T(x_0)\big)^2 \leq 2\big[(x-\theta_t(x_0))^2 + t^2 S_T^2(x_0)\big],$$

which after rearrangement yields $-(x-\theta_t(x_0))^2 \leq -\frac{1}{2}(x-x_0)^2 + t^2 S_T^2(x_0)$. Finally, we substitute this bound in (3.3) and find:

$$p_t(x_0,x) \leq \frac{\overline{\Lambda}_1}{t^{\frac{1}{2}}} \exp\left[-\overline{\Lambda}_2 \frac{(x-\theta_t(x_0))^2}{t}\right] \leq \frac{\overline{\Lambda}_1}{t^{\frac{1}{2}}} \exp\left[\overline{\Lambda}_2 T S_T^2(x_0)\right] \exp\left[-\overline{\Lambda}_2 \frac{(x-x_0)^2}{2t}\right] =: \frac{\Lambda_1}{t^{\frac{1}{2}}} \exp\left[-\Lambda_2 \frac{(x-x_0)^2}{t}\right],$$

where the constants Λ_1, Λ_2 being respectively $\Lambda_2 = \overline{\Lambda}_2/2$ and $\Lambda_1 = \overline{\Lambda}_1 e^{\overline{\Lambda}_2 T S_T^2(x_0)}$ are dependent on T but independent of t and x. □

We can now state the global assumption that we will consider on the density p_t of X_t, where in addition we provide a second Kusuoka-Stroock bound on the space derivatives of p_t. As a matter of clarity and relevance we do not provide a proof of this bound, however it can be found in Menozzi et al. [11, Theorem 1.2] for the interested reader.

Assumption 3.0.2. *For any $t \in (0,T]$, there exists $\beta \in \mathbb{N}^*$ such that the distribution of X_t has a β-times continuously differentiable density $p_t(x_0,\cdot)$. Moreover, for all $x \in \mathbb{R}$:*

$$0 < p_t(x_0,x) \leq \frac{\Lambda_1}{t^{\frac{1}{2}}} \exp\left[-\Lambda_2 \frac{(x-x_0)^2}{t}\right] \tag{3.4}$$

and

$$\left|\partial_x^k p_t(x_0,x)\right| \leq \frac{\Lambda_3(k)}{t^{q(k)}} \exp\left[-\Lambda_4(k) \frac{(x-x_0)^2}{t}\right], \quad \forall k \in \{1,\dots,\beta\}, \tag{3.5}$$

for some positive constants $\Lambda_1, \Lambda_2, \Lambda_3(k), \Lambda_4(k), q(k)$ dependent on T but independent of t and x, where we have made a dependence on k explicit in the last three constants.

This Assumption tells us that we will assume that the density and its space derivatives have Gaussian tails. Of course it should be expected that SDEs having Gaussian density should have respective Gaussian tails. Before stating the key result of this section, namely a Nikol'skii type condition on the density f, let us present an explicit example satisfying Assumption (3.0.2) for completeness.

Example 3.0.3 (Ornstein-Uhlenbeck process starting at zero)**.** *In this example, we show that the density of the Ornstein-Uhlenbeck process satisfies Assumption (3.0.2). To this end, we consider a process X_t solution to the following Langevin equation,*

$$X_t = -\beta \int_0^t X_s ds + \sigma W_t,$$

where $\beta > 0$, $\sigma > 0$ with W_t a Brownian motion and, for simplicity, an initial condition $X_0 = 0$. Then, upon introducing the integrating factor $e^{\beta t}$, the solution to the above SDE is

$$X_t = \sigma \int_0^t e^{-\beta(t-s)} dW_s.$$

Remark that we have a stochastic integral of a deterministic function with respect to a Gaussian process, hence the distribution of X_t is Gaussian. Furthermore it is not difficult to see that, given $X_0 = 0$, $\mathbb{E}(X_t) = 0$ and using

the Itô isometry we find

$$\mathrm{Var}(X_t) = \mathbb{E}\left[\left(\sigma\int_0^t e^{-\beta(t-s)}dW_s\right)^2\right] = \sigma^2\mathbb{E}\left(\int_0^t e^{-2\beta(t-s)}ds\right) = \frac{\sigma^2}{2\beta}\big(1-e^{-2\beta t}\big).$$

As a result, given that $X_0 = 0$, for all $t \in (0,T]$ we find $X_t \sim \mathcal{N}\big(0, \frac{\sigma^2}{2\beta}\big(1-e^{-2\beta t}\big)\big)$. The density of X_t is thus

$$p_t(0,x) = \frac{1}{\sqrt{\pi\frac{\sigma^2}{\beta}\big(1-e^{-2\beta t}\big)}}\exp\left(-\frac{x^2}{\frac{\sigma^2}{\beta}\big(1-e^{-2\beta t}\big)}\right).$$

Let us now find a bound on p_t, i.e. find some constants Λ_1 and Λ_2. Remark that the function $u \mapsto (1-e^{-u})/u$ is decreasing on $(0,\infty)$ and thus $1-e^{-2\beta t} \geq (t/T)\big(1-e^{-2\beta T}\big)$ for all $t \in (0,T]$. On the other hand, upon a first order Taylor expansion of the exponential we have $e^{-2\beta t} \geq 1-2\beta t$. Therefore, it is straightforward to verify that we may bound $p_t(0,x)$ as follows,

$$0 < p_t(0,x) \leq \frac{1}{t^{\frac{1}{2}}\sqrt{\pi\frac{\sigma^2}{\beta T}\big(1-e^{-2\beta T}\big)}}\exp\left(-\frac{x^2}{2\sigma^2 t}\right),$$

where the constants $\Lambda_1 := \big[\pi(\sigma^2/(\beta T))\big(1-e^{-2\beta T}\big)\big]^{-1/2}$ and $\Lambda_2 := (2\sigma^2)^{-1}$ are indeed dependent on T but independent of t and x. We will not delve into the tedious calculations for the k-th partial derivative of $p_t(0,x)$, however we remark the following: taking successive derivatives of the upper-bound on $p_t(0,x)$, $|\partial_x^k p_t(0,x)|$ will be bounded by a term of the form

$$\left|\partial_x^k C(t^{1/2})\exp\left(-\frac{\tilde{C}x^2}{t}\right)\right| \sim \left|C(t^{1/2})\left(\frac{2\tilde{C}}{t}\right)^{k/2}\big(\mathrm{He}_k(x,t)\big)\exp\left(-\frac{\tilde{C}x^2}{t}\right)\right|,$$

where $\mathrm{He}_k(x,t)$ is the k-th Hermite polynomial (i.e. a polynomial of order k), and $C(t^{1/2}), \tilde{C}$ are the constants from the upper-bound on $p_t(0,x)$. Thus, we recover a global constant that will be dependent on k and we will also have a term of the form $t^{-q(k)}$ for some constant $q(k)$ dependent on k (in fact choosing $q(k) = (k+1)/2$ works in this case). Finally one can verify that, for any $k \in \{1,\ldots,\beta\}$, the product of a Hermite polynomial of order k with $\exp(-\tilde{C}x^2/t)$ may be bounded by some constant $C_k > 0$ (indeed, the quantity $|x|^k\exp(-\tilde{C}x^2/t)$ is bounded for all k). We thus recover a term of the order $C_k\exp(\mathcal{O}(-x^2/t))$ which, combined with $t^{-q(k)}$, is exactly the bound that we were expecting in (3.5).

For the purpose of the study of the risk bounds that we will see later on, we now provide a Nikol'skii type condition on the density $f(x)$ (see [12, pp.5]). This particular bound gives us a way to control the forward/backward difference with respect to a parameter θ of the k-th derivative of f.

Lemma 3.0.4. *Under Assumptions (4.1.1) and (3.0.2), for the density $f(x) > 0$ there exists a constant $\Lambda_5 > 0$ depending on T but not on t_0, such that for every $k \in \{0,\ldots,\beta-1\}$, $x \in \mathbb{R}$ and $\theta \in \mathbb{R}$,*

$$\int_{\mathbb{R}}\Big(f^{(k)}(x+\theta) - f^{(k)}(x)\Big)^2 dx \leq \frac{\Lambda_5}{t_0^{2q(k+1)}}\big(\theta^2 + |\theta|^3\big). \tag{3.6}$$

Proof. First, by Assumption (3.0.2), since $p_t(x_0,x) > 0$ for all $(t,x) \in (0,T]\times\mathbb{R}$, it is clear that

$$f(x) = \frac{1}{T-t_0}\int_{t_0}^T p_t(x_0,x)dt > 0.$$

We now show the main statement. The idea will be to prove the statement twice, once for the forward

difference ($\theta \in \mathbb{R}^+$) and the other for the backward difference ($\theta \in \mathbb{R}^-$). Furthermore, for simplicity and without loss of generality, since the bound holds for all $x \in \mathbb{R}$ we will consider the shift $x \mapsto x + x_0$ in (3.6) so that we do not carry the x_0 in the upper-bound (3.5). To this end, set $\theta \in \mathbb{R}^+$ and $k \in \{0, \dots, \beta - 1\}$ to find,

$$\begin{aligned}
\|f^{(k)}(\cdot+\theta) - f^{(k)}(\cdot)\|_2^2 &:= \int_{\mathbb{R}} \Big(f^{(k)}(x+x_0+\theta) - f^{(k)}(x+x_0) \Big)^2 dx \\
&= \frac{1}{(T-t_0)^2} \int_{\mathbb{R}} \left[\int_{t_0}^{T} \big(\partial_x^k p_t(x_0, x+x_0+\theta) - \partial_x^k p_t(x_0, x+x_0) \big) dt \right]^2 dx \\
&\leq \frac{1}{T-t_0} \int_{t_0}^{T} \int_{\mathbb{R}} \big(\partial_x^k p_t(x_0, x+x_0+\theta) - \partial_x^k p_t(x_0, x+x_0) \big)^2 dx dt,
\end{aligned}$$

where this last inequality holds by Cauchy-Schwarz. Now, for any $\theta \in \mathbb{R}^+$, by the Mean Value Theorem we know that there exists a constant $c \in [x, x+\theta]$ such that $\theta p_t'(x_0, c+\theta) = p_t(x_0, x+x_0+\theta) - p_t(x_0, x+x_0)$. Applying this fact for the k-th derivative and using (3.5), we may further upper-bound as follows,

$$\begin{aligned}
\|f^{(k)}(\cdot+\theta) - f^{(k)}(\cdot)\|_2^2 &\leq \frac{\theta^2}{T-t_0} \int_{t_0}^{T} \int_{\mathbb{R}} \sup_{z \in [x, x+\theta]} \big| \partial_x^{k+1} p_t(x_0, z+x_0) \big|^2 dx dt \\
&\leq \Lambda_3 (k+1)^2 \frac{\theta^2}{T-t_0} \int_{t_0}^{T} \frac{1}{t^{2q(k+1)}} \int_{\mathbb{R}} \sup_{z \in [x, x+\theta]} \underbrace{\exp\left(-2\Lambda_4 (k+1) \frac{z^2}{t} \right)}_{=: \lambda(z)} dx dt.
\end{aligned}$$

The final step is to decompose the integral over $x \in \mathbb{R}$ to explicitly find a supremum of the function on different intervals. Note that the exponential function above has the form of a bell curve, where it is straightforward to verify that it attains a maximum value at $z = 0$ in its concave region over $z \in \big(-\sqrt{t/\Lambda_4(k+1)}, \sqrt{t/\Lambda_4(k+1)} \big)$. We may thus choose to decompose this final integral over the regions $(-\infty, -\theta)$, $[-\theta, 0]$ and $(0, \infty)$, such that the maximum is attained in the region $[-\theta, 0]$. As a result we find:

$$\begin{aligned}
\|f^{(k)}(\cdot+\theta) - f^{(k)}(\cdot)\|_2^2 &\leq \Lambda_3 (k+1)^2 \frac{\theta^2}{T-t_0} \int_{t_0}^{T} \frac{1}{t^{2q(k+1)}} \times \\
&\qquad \left[\int_{-\infty}^{-\theta} \sup_{z \in [x, x+\theta]} \lambda(z) dx + \int_{-\theta}^{0} \sup_{z \in [x, x+\theta]} \lambda(z) dx + \int_{0}^{\infty} \sup_{z \in [x, x+\theta]} \lambda(z) dx \right] dt \\
&= \Lambda_3 (k+1)^2 \frac{\theta^2}{T-t_0} \int_{t_0}^{T} \frac{1}{t^{2q(k+1)}} \times \\
&\qquad \left[\int_{-\infty}^{-\theta} \exp\left(-2\Lambda_4 (k+1) \frac{(x+\theta)^2}{t} \right) dx + \theta + \int_{0}^{\infty} \exp\left(-2\Lambda_4 (k+1) \frac{x^2}{t} \right) dx \right] dt,
\end{aligned}$$

where we have used the facts that the maximum of $\lambda(z)$ is attained at $z = x + \theta$ on the interval $(-\infty, -\theta)$ and at $z = x$ on the interval $(0, \infty)$. Of course, $\lambda(z) = 1$ over the interval $[-\theta, 0]$. Finally, upon a change of variable in the first integral, also using that $\exp(-2\Lambda_4(k+1)t^{-1}x^2) \leq \exp(-2\Lambda_4(k+1)T^{-1}x^2)$ and $t^{-2q(k+1)} \leq t_0^{-2q(k+1)}$ we conclude:

$$\begin{aligned}
\|f^{(k)}(\cdot+\theta) - f^{(k)}(\cdot)\|_2^2 &\leq \Lambda_3 (k+1)^2 \frac{\theta^2}{T-t_0} \int_{t_0}^{T} \frac{1}{t^{2q(k+1)}} \left[\theta + 2 \int_{0}^{\infty} \exp\left(-2\Lambda_4 (k+1) \frac{x^2}{t} \right) dx \right] dt \\
&\leq \frac{1}{t_0^{2q(k+1)}} \left(\gamma_1 \theta^2 + \theta^3 \max_{\xi \in \{0, \dots, \beta-1\}} \Lambda_3 (\xi+1)^2 \right),
\end{aligned}$$

where the constant γ_1 is given by,

$$\gamma_1 := 2 \max_{\xi \in \{0,\dots,\beta-1\}} \left\{ \Lambda_3(\xi+1)^2 \int_0^\infty \exp\left(-2\Lambda_4(k+1)\frac{x^2}{T}\right) dx \right\}.$$

For example, choosing $\Lambda_5 := \max\{\gamma_1, \max_\xi \Lambda_3(\xi+1)^2\}$ yields the result for the forward difference. In a very similar way with $\theta \in \mathbb{R}^+$ and $k \in \{0,\dots,\beta-1\}$, for the backward difference we find,

$$\begin{aligned}
\|f^{(k)}(\cdot-\theta) - f^{(k)}(\cdot)\|_2^2 &\leq \frac{\theta^2}{T-t_0} \int_{t_0}^T \int_{\mathbb{R}} \sup_{z\in[x-\theta,x]} \left|\partial_x^{k+1} p_t(x_0, z+x_0)\right|^2 dx dt \\
&\leq \Lambda_3(k+1)^2 \frac{\theta^2}{T-t_0} \int_{t_0}^T \frac{1}{t^{2q(k+1)}} \times \\
&\qquad \left[\int_{-\infty}^0 \exp\left(-2\Lambda_4(k+1)\frac{x^2}{t}\right) dx + \theta + \int_\theta^\infty \exp\left(-2\Lambda_4(k+1)\frac{(x-\theta)^2}{t}\right) dx \right] dt \\
&\leq \Lambda_3(k+1)^2 \frac{\theta^2}{T-t_0} \int_{t_0}^T \frac{1}{t^{2q(k+1)}} \left[\theta + 2\int_0^\infty \exp\left(-2\Lambda_4(k+1)\frac{x^2}{t}\right) dx \right] dt \\
&\leq \frac{1}{t_0^{2q(k+1)}} \left(\gamma_1 \theta^2 + \theta^3 \max_{\xi\in\{0,\dots,\beta-1\}} \Lambda_3(\xi+1)^2 \right) \\
&\leq \frac{\Lambda_5}{t_0^{2q(k+1)}} \left(\theta^2 + |\theta|^3\right).
\end{aligned}$$

□

Remark 3.0.5 (Choice of t_0)**.** *As we will see, the given assumptions and the Nikol'skii condition on the density f are crucial results in this paper, but we have to be careful when choosing t_0 in order to obtain reasonable risk-bounds. Indeed, since the Kusuoka-Stroock bounds and its derivatives explode for small values of t_0 one has to adapt the time interval in practice. For instance, one could take t_0 to be quite large (e.g. larger than 1) or consider T to be as small as possible. For completeness, the authors in [12] also provide risk-bounds which are independent of t_0 for the case where $T > 1$.*

4 | Nonparametric Kernel Estimation of the Drift Function

In this chapter, we deal with the nonparametric Nadaraya-Watson estimator of the drift function computed from independent continuous observations of a stochastic differential equation (SDE). In particular, for $b, \sigma : \mathbb{R} \to \mathbb{R}$ two continuous functions and $(W_t)_{t\in[0,T]}$ a Brownian motion, we consider the following SDE:

$$X_t = x_0 + \int_0^t b(X_s)ds + \int_0^t \sigma(X_s)dW_s, \quad t \in [0,T]. \tag{4.1}$$

We will provide a nonparametric estimator of the drift function $b(X_t)$ from $N \in \mathbb{N}^*$ i.i.d. copies $\{X_t^i\}_{i=1}^N$ solving (4.1) and from repeated continuous observations of the sample paths. We thus consider the following family of SDEs for N i.i.d. copies of a Brownian motion $\{W^1, \dots, W^N\}$:

$$X_t^i = x_0 + \int_0^t b(X_s^i)ds + \int_0^t \sigma(X_s^i)dW_s^i, \quad t \in [0,T] \text{ and } i \in \{1,\dots,N\}. \tag{4.2}$$

We then consider the access to the N random paths through the noisy observations $\{Y^i\}_{i=1}^N$ given by

$$Y^i = b(X^i) + \xi_i, \quad i \in \{1,\dots,N\}, \tag{4.3}$$

such that the Nadaraya-Watson estimation concerns the estimation of the measurable function b giving an estimator $\widehat{b}$, where $\xi_1, \dots, \xi_N$ are N i.i.d. centred random variables with a strictly positive variance and independent of the SDE samples $X^1, \dots, X^N$. We note that the diffusion coefficient can be almost surely determined by the process, i.e. it can be estimated without any error through its continuous observation throughout a time interval [1].

4.1 Nadaraya-Watson Estimator of the Drift Function

In this section, we motivate the construction of the Nadaraya-Watson estimator for the drift function starting from the density $p_t(x_0, \cdot)$ of the diffusion process X_t, with $t \in [0,T]$. As we will shortly see, the estimator of the drift will differ slightly from the general form of a Nadaraya-Watson estimator given in Proposition (2.0.4). For the remainder of this paper, let us suppose that the density $p_t(x_0, x)$ exists, is well-defined and that appropriate regularity conditions are satisfied (see Section (3) for the details). We furthermore make the usual assumption on the existence and uniqueness of a strong solution of the SDE, namely a Lipschitz condition on b and σ.

Assumption 4.1.1 (Lipschitz condition)**.** *There exists a positive constant C such that, for all $x, y \in \mathbb{R}$:*

$$|b(x) - b(y)| + |\sigma(x) - \sigma(y)|_F \leq C|x - y|. \tag{4.4}$$

Now, since $t \mapsto p_t(x_0, x)$ belongs to $L^1([0,T])$, we may define the function $f(x)$ for all $x \in \mathbb{R}$ as:

$$f(x) := \frac{1}{T-t_0}\int_{t_0}^{T} p_t(x_0,x)dt, \qquad \forall t_0 > 0. \tag{4.5}$$

Since $p_t(x_0,x)$ represents the probability density that the process X_t is at a point $x \in \mathbb{R}$ at time t, this function measures how much time on average the process X_t spends at x in the time-interval $[t_0,T]$. We note that f is a well-defined density function since,

$$\int_{\mathbb{R}} f(x)dx = \frac{1}{T-t_0}\int_{t_0}^{T}\int_{\mathbb{R}} p_t(x_0,x)dxdt = \frac{1}{T-t_0}\int_{t_0}^{T} dt = 1.$$

Let us briefly justify why one would consider the density (4.5) in our setup. Since we are considering the continuous observation of samples paths of time-homogeneous stochastic differential equations, an advantage is that we can construct density estimators in terms of the occupation-time density of the process [13]. As a result, for every sample path we may consider the continuous empirical measure to be given by the normalized occupation-time density

$$\mu_T^{(i)}(x) := \frac{1}{T}\int_0^T \mathbb{1}_{\{X_t^i = x\}}dt.$$

Then since we consider N i.i.d. sample paths of the SDE, the density of interest for the estimators is given by the empirical measure $\mu_{T,N}(x) := \frac{1}{N}\sum_{i=1}^N \mu_T^{(i)}(x)$ for which on average precisely yields the density function $\frac{1}{T}\int_0^T p_t(x_0,x)dt$. Note also that this density is defined on $[t_0,T]$ rather than $[0,T]$ since the Kusuoka-Stroock bounds on the density p_t explode as t approaches zero and are not integrable on $(0,T]$. This is a subtle technicality, however essential since these bounds will control the variance and bias terms in the later sections.

Remark 4.1.2 (Ergodicity)**.** *We note that considering densities in the form of (4.5) has a significant advantage when considering SDEs with an ergodicity assumption. Indeed, as mentioned in [14], one can show that there exists an $x_0 \in \mathbb{R}$ such that the sequence of measures $\mu_T(\cdot) := \frac{1}{T}\int_0^T p_s(x_0,\cdot)dt$ is tight (this heuristically guarantees that the sequence of measures does not escape to infinity). Thus, using Prohorov's Theorem, we may conclude that there exists a subsequence of times $\{T_n\}_n$ such that $\{\mu_{T_n}\}_n$ weakly converges to some invariant measure μ for the transition density. In particular, for $\mathbb{E}_\mu|f| < \infty$, this gives an ergodicity condition given by*

$$\lim_{T\to\infty}\frac{1}{T}\int_0^T f(X_s)ds = \mathbb{E}_\mu(f) := \int_{\mathbb{R}} f(x)\mu(x)dx.$$

Then a specific advantage is that, for $T \gg 1$, one can use the measure $\mu_{T,f} := \frac{1}{T}\int_0^T f(X_s)ds$ as an estimator for $\mathbb{E}_\mu(f)$ and thus greatly simplifies the estimation procedure.

We furthermore consider a kernel $K : \mathbb{R} \to \mathbb{R}$ and set $K_h(x) := h^{-1}K(h^{-1}x)$. Let us define an estimator for the drift such that we first aggregate the N i.i.d. SDE sample paths after which we perform the drift estimation. To this end, we define the (continuous-time) Nadaraya-Watson estimator of the drift function b to be

$$\widehat{b}_{N,h}(x) = \frac{\widehat{bf}_{N,h}(x)}{\widehat{f}_{N,h}(x)} := \frac{\left(\frac{1}{N(T-t_0)}\sum_{i=1}^N\int_{t_0}^T K_h(X_t^i - x)dX_t^i\right)}{\left(\frac{1}{N(T-t_0)}\sum_{i=1}^N\int_{t_0}^T K_h(X_t^i - x)dt\right)}. \tag{4.6}$$

Let us now make some comments on the construction of this estimator, where we first explain the estimator $\widehat{f}_{N,h}(x)$ of $f(x)$. The quantity in the denominator has simply been obtained by replacing the density of X_t by its Parzen-Rosenblatt estimator counterpart (2.6). Indeed, if we replace $p_t(x_0,\cdot)$ by $\hat{p}_t^P(x_0,\cdot)$ in $f(x)$, we obtain

the following estimator for the marginal density:

$$\widehat{f}_{N,h}(x) = \frac{1}{T-t_0}\int_{t_0}^{T} \hat{p}_t^P(x_0,x)dt = \frac{1}{T-t_0}\int_{t_0}^{T} \frac{1}{Nh}\sum_{i=1}^{N} K\left(\frac{X_t^i - x}{h}\right)dt = \frac{1}{N(T-t_0)}\sum_{i=1}^{N}\int_{t_0}^{T} K_h(X_t^i - x)dt.$$

Let us now provide a possible explanation for the construction of the estimator in the numerator. We follow the general construction of the Nadaraya-Watson estimator, where its numerator is derived as in (2.7). In particular, we consider a two-dimensional transition density for which the Parzen-Rosenblatt estimator (2.6) of this joint density yields

$$\widehat{bf}_{N,h}(x) = \frac{1}{Nh(T-t_0)}\int_{t_0}^{T}\sum_{i=1}^{N} K\left(\frac{X_t^i - x}{h}\right)\Bigg(\underbrace{\frac{1}{h}\int_{\mathbb{R}} yK\left(\frac{Y_t^i - y}{h}\right)dy}_{(*)}\Bigg)dt.$$

Now, as we have seen previously in Remark (2.0.5), we already know that $(*)$ equals Y_t^i where we will shortly explicitly construct this process. For the time being, the previous expression reduces to

$$\widehat{bf}_{N,h}(x) = \frac{1}{Nh(T-t_0)}\sum_{i=1}^{N}\int_{t_0}^{T} K\left(\frac{X_t^i - x}{h}\right)Y_t^i dt = \frac{1}{N(T-t_0)}\sum_{i=1}^{N}\int_{t_0}^{T} K_h\left(X_t^i - x\right)Y_t^i dt.$$

At this point we see that we are almost done if from now we assume that $dX_t^i = Y_t^i dt$. Let us then heuristically motivate this assumption. We choose the construction for Y_t to be in the form of a regression-type model $Y_t := b(X_t) + \sigma(X_t)\dot{W}_t$. Of course, this is only a heuristic representation since we are taking the time-derivative of a Brownian motion which is by definition almost surely nowhere differentiable. In fact, Y_t rigorously makes sense only in an Itô integral form for which we recover the SDE model X_t. However, we note that Y_t may still be interpreted at a formal level by viewing $\dot{W}_t$ as Gaussian white noise such that $\mathbb{E}(\dot{W}_t) = 0$ and $\mathbb{E}(\dot{W}_t\dot{W}_s) = \delta(t-s)$. As a result, the only major assumption we are making when considering $dX_t = Y_t dt$ is that we are giving a specific form to the additive error term of the regression model, where Y_t may be written as $Y_t = b(X_t) + \epsilon_t$ for some white-noise-dependent error ϵ_t. This error is represented by the stochastic integral of the SDE and is such that: (†) for any adapted process (φ_t) satisfying $\mathbb{E}\big(\int_{t_0}^{T}\varphi_s^2\sigma(X_s)^2 ds\big) < \infty$ we have that $\mathbb{E}\big(\int_{t_0}^{T}\varphi_t\sigma(X_t)dW_t\big) = 0$. This justification might seem quite unsatisfactory since a regression model does not usually assume any specific construction for the error term, however note that the only assumption we have made is that we are considering centred errors. Then, the (†) property plays the role of the classical zero-mean error assumption in regression and overall justifies the construction of the estimator

$$\widehat{bf}_{N,h}(x) = \frac{1}{N(T-t_0)}\sum_{i=1}^{N}\int_{t_0}^{T} K_h\left(X_t^i - x\right)dX_t^i.$$

Remark 4.1.3. *One has to be careful when defining Gaussian white-noise since it is not a stochastic process indexed on time. To obtain a rigorous definition of Gaussian white noise, one has to turn to the concept of a generalized stochastic process and is properly defined through a linear functional on a space of test functions. The two main ingredients are given by the facts that a continuous function F may be viewed as a distribution via the mapping $(F,f) := \int_0^\infty F(t)f(t)dt$ and also by defining the notion of the distributional derivative of F which is just another distribution F' such that paired with a test function f yields $(F',f) = -(F,f')$. Then, considering a Brownian motion W as the random function, Gaussian white noise X is the distributional derivative such that once paired with a test function f satisfies $(X,f) = -(W,f') = -\int_0^\infty W(t)f'(t)dt$. Full details of the theory may be found in Kazimierz Sobczyk's book on Stochastic Differential Equations [15, pp. 60-63].*

Although we have given a slightly different construction, we will still verify that equation (4.6) is a suitable

estimator for the drift even if we have added a stochastic term. To this end, we follow the steps in [16, pp. 4] and fill in the details. We will (roughly) show that $\widehat{b}_{N,h}(x)$ seems to be an appropriate estimator of $b(x)$. On the one hand,

$$\begin{aligned}\mathbb{E}\big(\widehat{f}_{N,h}(x)\big) := \int_{\mathbb{R}} \widehat{f}_{N,h}(\xi) p_s(x_0,\xi) d\xi &\overset{\text{iid}}{=} \frac{1}{T-t_0}\int_{t_0}^{T}\int_{\mathbb{R}} K_h(\xi-x) p_s(x_0,\xi) d\xi ds \\ &= \int_{\mathbb{R}} K_h(\xi-x)\left(\frac{1}{T-t_0}\int_{t_0}^{T} p_s(x_0,\xi) ds\right) d\xi \\ &= \int_{\mathbb{R}} K_h(\xi-x) f(\xi) d\xi \\ &= (K_h * f)(x) \xrightarrow{h\to 0} f(x),\end{aligned}$$

where we have used the fact that K_h is symmetric (i.e. $K_h(u) = K_h(-u)$, $\forall u$) and $*$ denotes the convolution operator. This final convergence result is a widely known fact from Real Analysis, for which the reader may refer to [17] for more details. On the other hand, using the fact that the expectation value of a stochastic integral is zero, we obtain:

$$\begin{aligned}\mathbb{E}\big(\widehat{bf}_{N,h}(x)\big) \overset{\text{iid}}{=} \frac{1}{T-t_0}\int_{t_0}^{T}\int_{\mathbb{R}} b(\xi) K_h(\xi-x) p_s(x_0,\xi) d\xi ds &= \int_{\mathbb{R}} K_h(\xi-x) b(\xi)\left(\frac{1}{T-t_0}\int_{t_0}^{T} p_s(x_0,\xi) ds\right) d\xi \\ &= \int_{\mathbb{R}} K_h(\xi-x) b(\xi) f(\xi) d\xi \\ &= \int_{\mathbb{R}} K_h(\xi-x)(bf)(\xi) d\xi \\ &= (K_h * (bf))(x) \xrightarrow{h\to 0} (bf)(x) = b(x)f(x),\end{aligned}$$

where $(bf)(x)$ denotes the pointwise product of $b(x)$ and $f(x)$. Upon taking a ratio of these two expressions we recover the drift function $b(x)$, which indicates that $\widehat{b}_{N,h}(x)$ should be a suitable estimator of $b(x)$.

4.2 Risk-Bounds on the Nadaraya-Watson Estimator of the Drift

The next stop in our study of the Nadaraya-Watson estimator of the drift function is the derivation of the so-called risk-bounds. In the sequel, we aim to derive those bounds and show that we recover the classical results of nonparametric regression. The approach will be made in three steps: first we derive two risk-bounds on the quantities in the numerator and denominator of (4.6), namely $\widehat{f}_{N,h}$ and $\widehat{bf}_{N,h}$, after which we will be able to conclude on a global risk-bound of the estimator. Interestingly, and perhaps surprisingly, it will turn out that the risk-bound of $\widehat{b}_{N,h}$ can be controlled by the sum of those of $\widehat{f}_{N,h}$ and $\widehat{bf}_{N,h}$ up to multiplicative constants. While the results and proofs in this section follow *Marie & Rosier* [12], we supplement their arguments with further computational details and expanded explanations in Appendix (A.1). We only provide a sketch of the proofs for the bounds in this section, prioritising the overall procedure over the mathematical details.

Assumption 4.2.1. *We consider a kernel K to be symmetric, continuous and belongs to $L^2(\mathbb{R}, dx)$.*

Assumption 4.2.2. *Suppose that there exists $v \in \mathbb{N}^*$ such that:*

$$\int_{\mathbb{R}} \big|z^{v+1} K(z)\big| dz < \infty \quad \text{and} \quad \int_{\mathbb{R}} z^k K(z) dz = 0, \ \ \forall k \in \{1, ..., v\}. \tag{4.7}$$

Remark 4.2.3. *See [9, Proposition 2.10] for the construction of kernels satisfying Assumption (4.2.2).*

The crucial first step that will be used in the risk bounds is the **bias-variance decomposition**. For $\widehat{g}(x)$ an estimator of a density function $g(x)$, the decomposition of the L^2-risk is derived as follows:

$$\begin{aligned}\mathbb{E}\big(\|\widehat{g}-g\|_2^2\big) := \mathbb{E}\left(\int_{\mathbb{R}}\big(\widehat{g}(x)-g(x)\big)^2dx\right) &= \mathbb{E}\left(\int_{\mathbb{R}}\big(\widehat{g}(x)-\mathbb{E}(\widehat{g}(x))\big)^2dx\right)+\mathbb{E}\left(\int_{\mathbb{R}}\big(\mathbb{E}(\widehat{g}(x))-g(x)\big)^2dx\right)\\ &= \int_{\mathbb{R}}\mathbb{E}\Big[\big(\widehat{g}(x)-\mathbb{E}(\widehat{g}(x))\big)^2\Big]dx+\int_{\mathbb{R}}\big(\mathbb{E}(\widehat{g}(x))-g(x)\big)^2dx \quad \rightsquigarrow \text{Fubini}\\ &= \int_{\mathbb{R}}\mathbb{V}_{\hat{g}}(x)dx+\int_{\mathbb{R}}\mathtt{bias}^2_{\hat{g}}(x)dx.\end{aligned}$$

Let us now start the study of the risk bounds, where we first state and sketch the proof of a bound on the estimator $\widehat{f}_{N,h}(x)$ of $f(x)$.

Proposition 4.2.4. *Under Assumptions (3.0.2), (4.1.1), (4.2.1) and (4.2.2) with $v=\beta$ we have,*

$$\mathbb{E}\left(\|\widehat{f}_{N,h}-f\|_2^2\right) \le \Lambda_6(t_0)h^{2\beta}+\frac{\|K\|_2^2}{Nh} \tag{4.8}$$

where

$$\Lambda_6(t_0) := \frac{\Lambda_5}{|(\beta-2)!|^2 t_0^{2q(\beta)}}\left(\int_{\mathbb{R}}|z|^\beta\big(1+|z|^{\frac{1}{2}}\big)|K(z)|dz\right)^2. \tag{4.9}$$

For the following sketch, we will prove the bound on the integrated variance and the bound for the integrated squared-bias only in the case where $\beta=1$. Full details of the proof along with the general case $\beta\ge 2$ may be found in Appendix (A.1).

Sketch of Proof.

• The general methodology is the following: we use the decomposition of the L^2-risk into the bias and the variance of $\widehat{f}_{N,h}$. To this end we have:

$$\mathbb{E}\left(\|\widehat{f}_{N,h}-f\|_2^2\right) = \int_{\mathbb{R}}\mathtt{bias}^2_{\hat{f}}(x)dx+\int_{\mathbb{R}}\mathbb{V}_{\hat{f}}(x)dx.$$

For the variance term, the details mainly rely on the facts that $\mathrm{Var}(X)\le\mathbb{E}(X^2)$ and also that the kernel K is symmetric. Concerning the squared-bias, it will become clear that we should use the Nikol'skii type condition on the density $f(x)$, and to this end the key step will be to first exchange the integral over $[t_0,T]$ in time (from the definition of f) to an integral over $\mathbb{R}$ in space.

• Let us present some of the details, where we first concentrate on the variance term. The first step is to provide a (crude) upper-bound directly from the definition of the variance. Indeed, for any random variable X, since $\mathbb{E}(X)^2\ge 0$ we conclude that $\mathrm{Var}(X) := \mathbb{E}(X^2)-\mathbb{E}(X)^2\le\mathbb{E}(X^2)$. Recall that we are considering $X^1,\dots,X^N$ i.i.d. copies of X and thus,

$$\begin{aligned}\mathbb{V}_{\hat{f}}(x) = \mathrm{Var}\left(\frac{1}{N(T-t_0)}\sum_{i=1}^N\int_{t_0}^T K_h(X_t^i-x)dt\right) &\overset{\text{iid}}{=} \frac{1}{N(T-t_0)^2}\mathrm{Var}\left(\int_{t_0}^T K_h(X_t-x)dt\right)\\ &\le \frac{1}{N}\mathbb{E}\left[\left(\frac{1}{T-t_0}\int_{t_0}^T K_h(X_t-x)dt\right)^2\right].\end{aligned}$$

Now, we use Cauchy-Schwarz to further upper-bound the above expression and use Fubini's Theorem to exchange integration and expectation. The remaining step will be to explicitly input the definition of the expectation under the density $f(x)$ so that we end up with an integration over the space $\mathbb{R}$ rather than time:

$$\mathbb{V}_{\hat{f}}(x) \leq \frac{1}{N(T-t_0)} \int_{t_0}^{T} \mathbb{E}(K_h(X_t - x)^2) dt = \frac{1}{N(T-t_0)} \int_{t_0}^{T} \int_{\mathbb{R}} K_h(z-x)^2 p_t(x_0, z) dz dt$$
$$= \frac{1}{N} \int_{\mathbb{R}} K_h(z-x)^2 f(z) dz.$$

By assumption, K is symmetric and thus upon a change of variable $v = (x-z)/h$ we obtain,

$$\int_{\mathbb{R}} \mathbb{V}_{\hat{f}}(x) dx \leq \frac{1}{N} \int_{\mathbb{R}} \int_{\mathbb{R}} K_h(z-x)^2 f(z) dz dx = \frac{\|K\|_2^2}{Nh}.$$

Let us now find a bound on the integrated squared-bias. Recall that $\texttt{bias}_{\hat{f}}(x) = \mathbb{E}\big(\widehat{f}_{N,h}(x)\big) - f(x)$ and also that we are considering $X^1, \dots, X^N$ i.i.d. copies of X. As for the variance, we again input the definition of the expectation under the density $f(x)$ and reduce our problem to an integration over space rather than time. Thus, upon a change of variable $w = (z-x)/h$:

$$\texttt{bias}_{\hat{f}}(x) = \mathbb{E}\left(\frac{1}{N(T-t_0)} \sum_{i=1}^{N} \int_{t_0}^{T} K_h(X_t^i - x) dt \right) - f(x)$$
$$\overset{\text{iid}}{=} \frac{1}{T-t_0} \int_{t_0}^{T} \mathbb{E}(K_h(X_t - x)) dt - f(x) = \int_{\mathbb{R}} K(w) \big(f(wh+x) - f(x) \big) dw,$$

where we have used the fact that the kernel K integrates to 1. From this point, one should recognize the form given in the Nikol'skii type condition on the density $f(x)$ and thus the idea will now be to use Lemma (3.0.4) on the integrated squared-bias. Let us consider here the case $\beta = 1$. By the Generalized Minkowski Inequality (see Lemma (A.0.2)), Lemma (3.0.4) applied with $\theta = hw$ and $k = 0$ gives us,

$$\int_{\mathbb{R}} \texttt{bias}_{\hat{f}}^2(x) dx = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} K(w) \big(f(wh+x) - f(x) \big) dw \right)^2 dx$$
$$\leq \left[\int_{\mathbb{R}} K(w) \left(\int_{\mathbb{R}} \big(f(hw+x) - f(x) \big)^2 dx \right)^{\frac{1}{2}} dw \right]^2 \qquad \rightsquigarrow \text{Minkowski}$$
$$\leq \left[\int_{\mathbb{R}} K(w) \left(\frac{\Lambda_5}{t_0^{2q(1)}} \big((hw)^2 + |hw|^3 \big) \right)^{\frac{1}{2}} dw \right]^2 \qquad \rightsquigarrow \text{Lemma (3.0.4)}$$
$$= h^2 \frac{\Lambda_5}{t_0^{2q(1)}} \left[\int_{\mathbb{R}} K(w) |w| (1 + |hw|)^{\frac{1}{2}} dw \right]^2 \leq \widehat{\Lambda}_6(t_0) h^2$$

where $\widehat{\Lambda}_6(t_0)$ is a bounded constant by Assumption (4.2.2) and is given by:

$$\widehat{\Lambda}_6(t_0) = \frac{\Lambda_5}{t_0^{2q(1)}} \left[\int_{\mathbb{R}} |K(w)| |w| \big(1 + |w|^{\frac{1}{2}} \big) dw \right]^2.$$

For $\beta \geq 2$ the steps are very similar than for the base case, where we write $f(wh+x) - f(x)$ as a Taylor expansion with integrated remainder. Upon two applications of the Generalized Minkowski Inequality and one of Lemma (3.0.4) the result follows. □

Remark 4.2.5. *We note that in the second to last equality of the integrated squared-bias we have chosen to extract $(hw)^2$ instead of $|hw|^3$. Indeed, since $h \in (0,1]$ it may be tempting to extract $|hw|^3$ instead of $(hw)^2$ to obtain a tighter risk bound. However, if h and w are ever both small then the risk-bound may get very large. A scenario that we avoid by only extracting a factor of $(hw)^2$.*

Let us find the optimal bandwidth to obtain the tightest bound possible in (4.8). Upon a derivative of the upper-

bound with respect to the bandwidth h, it is straightforward to verify that choosing $h_1 = \mathcal{O}\big(N^{-1/(2\beta+1)}\big)$ yields the tightest risk-bound. Also, for this specific choice h_1 the value attained by the risk-bound is of the order

$$\mathcal{O}\left(h_1^{2\beta} + \frac{1}{Nh_1}\right) \propto \mathcal{O}\left(N^{-\frac{2\beta}{2\beta+1}}\right),$$

matching the classical results from nonparametric regression. Recall that $\beta \geq 1$ is the order of differentiability from the Kusuoka-Stroock bounds on the density p_t. Then if we consider the scenario of having an infinitely differentiable density then the risk-bound cannot hope to be better than a rate approaching $\mathcal{O}(N^{-1})$. This order of convergence being a relatively slow rate from a computational point of view (which is not the point we are interested in here), this sublinear rate is to be expected from a practical point of view and we cannot hope to do better. Indeed, remark that for low values of β the corresponding rates are relatively worse than for a C^∞ density.

Proposition 4.2.6. *Under Assumptions (4.1.1) and (4.2.1) we have,*

$$\mathbb{E}\left(\|\widehat{bf}_{N,h} - bf\|_2^2\right) \leq \|(K_h * bf) - bf\|_2^2 + \frac{\Lambda_7(t_0)}{Nh} \tag{4.10}$$

where

$$\Lambda_7(t_0) := 2\|K\|_2^2\left(\|b\|_{2,f}^2 + \frac{1}{T-t_0}\|\sigma\|_{2,f}^2\right). \tag{4.11}$$

Sketch of Proof.

• The general methodology is the following: we use the decomposition of the L^2-risk into the bias and the variance of $\widehat{bf}_{N,h}$. To this end we have:

$$\mathbb{E}\left(\|\widehat{bf}_{N,h} - bf\|_2^2\right) = \int_{\mathbb{R}} \mathtt{bias}^2_{\hat{bf}}(x)dx + \int_{\mathbb{R}} \mathbb{V}_{\hat{bf}}(x)dx.$$

Now, the remaining steps will follow very closely the arguments used in the proof of Proposition (4.2.4). For the variance term, the computations again rely mainly on the observations that $\mathrm{Var}(X) \leq \mathbb{E}(X^2)$ and also that the kernel K is symmetric. As for the squared-bias, although the initial steps are identical, the conclusion differs slightly from the previous proof since an additional term appears due to the Itô Isometry. Nevertheless, the key idea will still be to first exchange the integral over $[t_0, T]$ in time (from the definition of f) to a spatial integral over $\mathbb{R}$.

• Let us present some details, where we first focus on the variance term. Recall that we are considering $X^1, \ldots, X^N$ i.i.d. copies of X, thus by using the fact that $\mathrm{Var}(X) \leq \mathbb{E}(X^2)$ and by explicitly substituting the SDE we find,

$$\begin{aligned}\mathbb{V}_{\hat{bf}}(x) = \mathrm{Var}\left(\frac{1}{N(T-t_0)}\sum_{i=1}^N \int_{t_0}^T K_h(X_t^i - x)dX_t^i\right) &\overset{\text{iid}}{=} \frac{1}{N(T-t_0)^2}\mathrm{Var}\left(\int_{t_0}^T K_h(X_t - x)dX_t\right) \\ &\leq \frac{2}{N(T-t_0)^2}\mathbb{E}\left[\left(\int_{t_0}^T K_h(X_t - x)b(X_t)dt\right)^2 + \left(\int_{t_0}^T K_h(X_t - x)\sigma(X_t)dW_t\right)^2\right].\end{aligned}$$

We may apply the Itô Isometry on the second term and also use Cauchy-Schwarz to bound the first term since $K \in L^2$ by assumption. Furthermore, by explicitly writing the expectation as an integral over the density $f(x)$ we conclude that

$$\mathbb{V}_{\hat{bf}}(x) \leq \frac{2}{N}\int_{\mathbb{R}} K_h(z-x)^2 b(z)^2 f(z)dz + \frac{2}{N(T-t_0)}\int_{\mathbb{R}} K_h(z-x)^2\sigma(z)^2 f(z)dz.$$

Moreover, by Assumption (4.2.1) we know that K is symmetric and $b, \sigma \in L^2(\mathbb{R}, f(x)dx)$. Thus, upon a change of variable, the integrated variance is bounded as follows:

$$\int_{\mathbb{R}} \mathbb{V}_{\hat{bf}}(x)dx \leq \frac{2\|K\|_2^2}{Nh}\left(\int_{\mathbb{R}} b(z)^2 f(z)dz + \frac{1}{T-t_0}\int_{\mathbb{R}} \sigma(z)^2 f(z)dz\right) = \frac{2\|K\|_2^2}{Nh}\left(\|b\|_{2,f}^2 + \frac{1}{T-t_0}\|\sigma\|_{2,f}^2\right).$$

Regarding the bias, we first recall that $\texttt{bias}_{\hat{bf}}(x) = \mathbb{E}\big(\widehat{bf}_{N,h}(x)\big) - b(x)f(x)$. As usual we use the fact that we are considering $X^1, \dots, X^N$ i.i.d. copies of X and thus,

$$\begin{aligned}\texttt{bias}_{\hat{bf}}(x) = \mathbb{E}\left(\frac{1}{N(T-t_0)}\sum_{i=1}^{N}\int_{t_0}^{T} K_h(X_t^i - x)dX_t^i\right) - b(x)f(x) &\overset{\text{iid}}{=} \frac{1}{T-t_0}\mathbb{E}\left(\int_{t_0}^{T} K_h(X_t - x)dX_t\right) - b(x)f(x) \\ &= (K_h * (bf))(x) - (bf)(x).\end{aligned}$$

Although we have omitted the details, we mention that we have used the fact that the Itô integral restricted to $\mathbb{H}^2$ is a martingale-valued map, and thus its expectation yields zero. Therefore, since f is bounded and b belongs to $L^2(\mathbb{R}, f(x)dx)$ we conclude:

$$\int_{\mathbb{R}} \texttt{bias}_{\hat{bf}}^2(x)dx = \|(K_h * bf) - bf\|_2^2.$$

□

Note that since $\mathbb{E}\big(\widehat{bf}_{N,h}(x)\big) = (K_h * bf)(x)$ then $\|(K_h * bf) - bf\|_2^2$ measures the quality of the choice of the kernel K_h and thus represents an *intrinsic* risk of the estimation procedure. Again for the choice of the optimal bandwidth minimizing the risk-bound of bf, *Marie & Rosier* [12] show that a bias-variance tradeoff of the order $h_2 = \mathcal{O}\big(N^{-1/(2\gamma+1)}\big)$ leading to a rate of $\mathcal{O}\big(N^{-2\gamma/(2\gamma+1)}\big)$ is attained by assuming the existence of a Nikol'skii type condition on bf. The assumption is that bf is $\gamma \in \mathbb{N}^*$ times continuously differentiable and that there exists $\varphi \in L^1(\mathbb{R}, |z|^{\gamma-1}K(z)dz)$ such that, for every $\theta \in \mathbb{R}$ and $h \in (0,1]$,

$$\int_{\mathbb{R}} \Big((bf)^{(\gamma-1)}(x+h\theta) - (bf)^{(\gamma-1)}(x)\Big)^2 dx \leq \varphi(\theta)h^2.$$

And moreover since K satisfies Assumption (4.2.2), with $v = \gamma$ we obtain $\|(K_h * bf) - bf\|_2^2 = \mathcal{O}(h^{2\gamma})$. Again by a differentiation with respect to h of the upper-bound we obtain the same rate $h_2 = \mathcal{O}\big(N^{-1/(2\gamma+1)}\big)$ yielding the tightest risk-bound. This specific choice of h_2 also yields the same value for the risk-bound, namely the rate

$$\mathcal{O}\left(h_2^{2\gamma} + \frac{1}{Nh_2}\right) \propto \mathcal{O}\left(N^{-\frac{2\gamma}{2\gamma+1}}\right).$$

We are now ready to combine the results and find a risk-bound for the global Nadaraya-Watson estimator.

Proposition 4.2.7. *Consider the (truncated) Nadaraya-Watson estimator*

$$\widehat{b}_{N,h}(x) := \frac{\widehat{bf}_{N,h}(x)}{\widehat{f}_{N,h}(x)}\mathbb{1}_{\left\{\widehat{f}_{N,h}(x) > \frac{m}{2}\right\}}, \tag{4.12}$$

where $h > 0$ and assume that $f(x) > m > 0$ for all $x \in [A,B]$ ($m \in (0,1]$ and $A, B \in \mathbb{R}$ such that $A < B$). Under Assumptions (3.0.2), (4.1.1), (4.2.1) and (4.2.2) with $v = \beta$ we have,

$$\mathbb{E}\left(\|\widehat{b}_{N,h} - b\|_{f,A,B}^2\right) \leq \frac{\Lambda_8}{m^2}\left[\|(K_h * bf) - bf\|_2^2 + \frac{\Lambda_7(t_0)}{Nh} + 2\left(\Lambda_6(t_0)h^{2\beta} + \frac{\|K\|_2^2}{Nh}\right)\right] \tag{4.13}$$

where

$$\Lambda_8 := 8\big(\|f\|_\infty \vee \|b^2 f\|_\infty\big). \tag{4.14}$$

Sketch of Proof. The strategy for this proof is to appropriately control the estimator $\widehat{f}_{N,h}$. For this, we introduce a benchmark $m>0$ in order to guarantee that $\widehat{f}_{N,h}$ does not get too small, otherwise we could have $\widehat{b}_{N,h}$ to be relatively large. We will consider that above the threshold of $m/2$, $\widehat{f}_{N,h}$ is small enough so as to make $\widehat{b}_{N,h}$ finite and of reasonable order. We may now decompose $\widehat{b}_{N,h}$ in the following way,

$$\begin{aligned}\widehat{b}_{N,h}-b &= \big(\widehat{b}_{N,h}-b\big)\mathbb{1}_{\left\{\widehat{f}_{N,h}(\cdot)>\frac{m}{2}\right\}}+\big(\widehat{b}_{N,h}-b\big)\mathbb{1}_{\left\{\widehat{f}_{N,h}(\cdot)\leq\frac{m}{2}\right\}}\\ &= \left[\frac{\widehat{bf}_{N,h}-bf}{\widehat{f}_{N,h}}+bf\left(\frac{1}{\widehat{f}_{N,h}}-\frac{1}{f}\right)\right]\mathbb{1}_{\left\{\widehat{f}_{N,h}(\cdot)>\frac{m}{2}\right\}}-b\mathbb{1}_{\left\{\widehat{f}_{N,h}(\cdot)\leq\frac{m}{2}\right\}},\end{aligned}$$

where we have written $\widehat{b}_{N,h}-b$ over $\left\{\widehat{f}_{N,h}(\cdot)>\frac{m}{2}\right\}$ in a convenient way for the risk-bounds, and used the fact that we force $\widehat{b}_{N,h}$ to be zero over $\left\{\widehat{f}_{N,h}(\cdot)\leq\frac{m}{2}\right\}$. The idea is now to bound each of the three terms separately. As explicitly shown in the details of the proof, it should be clear from the above decomposition that the very first term can be bounded by the risk bound $\mathbb{E}\big(\|\widehat{bf}_{N,h}-bf\|_2^2\big)$ of bf, while the last two terms are (almost) only dependent on the density f. Thus, we bound the risk of $(\widehat{b}_{N,h}-b)$ by a sum of the risk-bounds on bf and f as follows:

$$\mathbb{E}\big(\|\widehat{b}_{N,h}-b\|_{f,A,B}^2\big)\leq\frac{8\big(\|f\|_\infty\vee\|b^2f\|_\infty\big)}{m^2}\big(\mathbb{E}\big(\|\widehat{bf}_{N,h}-bf\|_2^2\big)+2\mathbb{E}\big(\|\widehat{f}_{N,h}-f\|_2^2\big)\big).$$

□

Once more we provide a choice h_3 of bandwidth for which the upper-bound is minimized. Note that this is now straightforward to find since the global estimator is the sum (up to multiplicative constants) of the risks of bf and f. As a result, since the risk-bound is of the order $\mathcal{O}\big(h^{2\beta}+h^{2\gamma}+2/(Nh)\big)$ and upon differentiation with respect to h, choosing

$$h_3=\mathcal{O}\left(N^{-\frac{1}{2(\beta\wedge\gamma)+1}}\right)$$

yields the tightest risk-bound. The corresponding value of the bound is of the order,

$$\mathcal{O}\left(h^{2\beta}+h^{2\gamma}+2/(Nh)\right)\propto\mathcal{O}\left(N^{-2\left[\left(\frac{\gamma}{2\gamma+1}\right)\wedge\left(\frac{\beta}{2\beta+1}\right)\right]}\right)=\mathcal{O}\left(N^{-\frac{2(\beta\wedge\gamma)}{2(\beta\wedge\gamma)+1}}\right).$$

Remark 4.2.8. *We note that a restriction to Proposition (4.2.7) is that m is unknown in general. As a result it must be replaced by an estimator and a common choice in practice is to consider the minimum of $\widehat{f}$ so that,*

$$\widehat{m}_{N,h}:=\min\big\{\widehat{f}_{N,h}(x):x\in[A,B]\big\}.$$

4.3 Methods for Bandwidth Selection

For completeness, we aim to briefly present some methods for bandwidth selection. In particular, we will focus on the so-called Leave-one-out Cross Validation method (LooCV) and the Penalized Comparision to Overfitting method (PCO).

Recall that we consider the standard regression noisy model $Y^i=b(X^i)+\xi_i$, where $i=\{1,\dots,N\}$, such that $(\xi_i)_{1\leq i\leq N}$ are i.i.d. centred errors and our aim was to estimate the drift function $b(\cdot)$ of an SDE from independent continuous observations of the sample paths. As we have mentioned in the previous section, an important task is to find the bandwidth h yielding the optimal bias-variance tradeoff. This tradeoff emanating from the risk-bounds of the previous section is all about balancing h and its inverse. Indeed, recall for instance that the risk of $\widehat{f}_{N,h}$ could be upper-bounded by $\Gamma_1h^{2\beta}+\Gamma_2/(Nh)$ for some constants Γ_1,Γ_2 and where $\beta\in\mathbb{N}^*$. This

bound was obtained by noting that the integrated squared-bias could be upper-bounded by $\Gamma_1 h^{2\beta}$ and the integrated variance by $\Gamma_2/(Nh)$. Thus, the upper-bounds on the integrated variance and squared-bias operate in different ways as the bandwidth h varies. Indeed, the bound on the integrated variance decreases as h gets large however the one of the integrated squared-bias increases. The specific choice of bandwidth h that optimizes these two quantities is the one achieving the best tradeoff. Note that if we choose too small of a bandwidth h this will give a relatively large integrated variance and thus will most likely lead to *undersmoothing*. On the contrary, choosing too large of a bandwidth will yield a relatively large integrated squared-bias and thus will most likely lead to *oversmoothing*. Let us then look at possible methods to choose this optimal bandwidth all developed in the work of *Marie & Rosier* [12].

4.3.1 Leave-one-out Cross Validation Method

We first start with the **Leave-one-out Cross Validation** method (LooCV) for which the intuition is to minimize an average of error terms. More precisely, we consider the true model Y^i and train the model using the Nadaraya-Watson estimator $\widehat{b}^{(-i)}$ where the $(-i)$ indicates the sample we are leaving out in order to train on the other $N-1$ samples. We then seek for the optimal bandwidth h to minimize the mean-squared error, i.e. an average of $\left\|Y^i - \widehat{b}^{(-i)}\right\|_2^2$. Then the idea is fairly simple. We first note that the Nadaraya-Watson estimator of the drift function may be written as a weighted representation since, from (4.6),

$$\widehat{b}_{N,h}(x) = \sum_{i=1}^{N} \int_{t_0}^{T} \omega_t^i(x) dX_t^i \qquad \text{where} \quad \omega_t^i := \frac{K_h(X_t^i - x)}{\sum_{j=1}^{N} \int_{t_0}^{T} K_h(X_t^j - x) dt}; \;\; \forall (t,i) \in [t_0, T] \times \{1, \dots, N\}.$$

Of course, the weights satisfy $\sum_{i=1}^{N} \int_{t_0}^{T} \omega_t^i(x) dt = 1$ and we set the leave-one-out estimator as

$$\widehat{b}_{N,h}^{(-i)}(x) := \sum_{j \in \{1,\dots,N\} \setminus \{i\}} \int_{t_0}^{T} \omega_t^j(x) dX_t^j; \qquad \forall i \in \{1, \dots, N\}.$$

Formally, since we do not have access to Y_t directly, the authors extend the LooCV method by assuming that $dX_t = Y_t dt$ only for computational purposes. This idea leads us to consider the regression-like model from the setup of this paper $Y_t = b(X_t) + \sigma(X_t)\dot{W}_t$ where $\dot{W}_t$ represents Gaussian white noise. Then, from the usual representation of the LooCV criterion we write,

$$\begin{aligned}
\text{LooCV}(h) := \sum_{i=1}^{N} \int_{t_0}^{T} \left(Y_t^i - \widehat{b}_{N,h}^{(-i)}(X_t^i)\right)^2 dt &= \sum_{i=1}^{N} \left(\int_{t_0}^{T} \widehat{b}_{N,h}^{(-i)}(X_t^i)^2 dt - 2 \int_{t_0}^{T} \widehat{b}_{N,h}^{(-i)}(X_t^i) Y_t^i dt \right) + \sum_{i=1}^{N} \int_{t_0}^{T} (Y_t^i)^2 dt \\
&= \sum_{i=1}^{N} \left(\int_{t_0}^{T} \widehat{b}_{N,h}^{(-i)}(X_t^i)^2 dt - 2 \int_{t_0}^{T} \widehat{b}_{N,h}^{(-i)}(X_t^i) dX_t^i \right) + \sum_{i=1}^{N} \int_{t_0}^{T} (Y_t^i)^2 dt \\
&\equiv \widetilde{\text{LooCV}}(h) + \underbrace{\sum_{i=1}^{N} \int_{t_0}^{T} (Y_t^i)^2 dt}_{(\Delta)}.
\end{aligned}$$

Then instead of working with LooCV(h) for which the term (Δ) is ill-defined, since Gaussian white noise only makes sense as a random distribution and not as a random variable, we can work with $\widetilde{\text{LooCV}}(h)$. Indeed, we remark that this undefined term is independent of h and thus for the LooCV method minimizing LooCV(h) is equivalent to minimizing $\widetilde{\text{LooCV}}(h)$ which is now an implementable problem since this term only depends on the SDE paths and the leave-one-out estimator. We note also that the problem of minimizing $\widetilde{\text{LooCV}}(h)$ is independent of the assumption that $dX_t = Y_t dt$.

Remark 4.3.1. *In practice, Marie & Rosier base this LooCV method on a discrete time approximate of the Nadaraya-Watson estimator, instead of the continuous version that we have presented above. For a dissection of $[t_0,T]$ using some timepoints $(t_0,t_1,\dots,t_n)$ where $t_j = t_0 + (T-t_0)(j/n)$ for every $j \in \{1,\dots,n\}$, they define the discrete-time estimator $\widehat{b}_{n,N,h}$ as,*

$$\widehat{b}_{n,N,h} = \frac{\widehat{bf}_{n,N,h}}{\widehat{f}_{n,N,h}} := \frac{\Big(\frac{1}{N(T-t_0)}\sum_{i=1}^{N}\sum_{j=0}^{n-1} K_h(X^i_{t_j}-x)\big(X^i_{t_j+1}-X^i_{t_j}\big)\Big)}{\Big(\frac{1}{nN}\sum_{i=1}^{N}\sum_{j=0}^{n-1} K_h(X^i_{t_j}-x)\big(t_{j+1}-t_j\big)\Big)}.$$

This construction is very similar in nature to the one we have investigated in (4.6), where instead of considering the integrated continuous process dX_t and time process dt, we look at sums of increments over the interval $[t_0,T]$ defined on the truncation $(t_0,t_1,\dots,t_n)$. We thus obtain a similar criterion to minimize in discrete times,

$$\widetilde{\mathrm{LooCV}}_n(h) = \sum_{i=1}^{N}\left(\sum_{j=0}^{n-1}\widehat{b}^{(-i)}_{n,N,h}(X^i_{t_j})^2(t_{j+1}-t_j) - 2\sum_{j=0}^{n-1}\widehat{b}^{(-i)}_{n,N,h}(X^i_{t_j})(X^i_{t_{j+1}}-X^i_{t_j})\right)$$

where, for all $(j,i) \in \{0,\dots,n-1\}\times\{1,\dots,N\}$, the quantities are now given by

$$\widehat{b}_{n,N,h}(x) = \sum_{i=1}^{N}\sum_{j=0}^{n-1}\omega^i_j(x)(X^i_{t_{j+1}}-X^i_{t_j}) \qquad \text{with} \quad \omega^i_j := \frac{K_h(X^i_{t_j}-x)}{\sum_{\alpha=1}^{N}\sum_{\beta=0}^{n-1}K_h(X^\alpha_{t_\beta}-x)(t_{\beta+1}-t_\beta)},$$

and where the leave-one-out estimator is now set as

$$\widehat{b}^{(-i)}_{n,N,h}(x) := \sum_{k\in\{1,\dots,N\}\setminus\{i\}}\sum_{j=0}^{n-1}\omega^k_j(x)(X^k_{t_j+1}-X^k_{t_j}); \quad \forall i \in \{1,\dots,N\}.$$

4.3.2 Penalized Comparison to Overfitting Method

We also mention that *Marie & Rosier* present a novel minimax approach based on the work of *Lacour, Massart & Rivoirard* [18] for which a comparison is made with the classical Lepski's pairwise comparison method consisting in choosing the optimal bandwidth h as,

$$\widehat{h} = \operatorname*{arg\,max}_{h\in\mathcal{H}}\big\{\|\widehat{f}_{h'}-\widehat{f}_h\|_2^2 \le V_n(h') \quad \text{for any } h' \in \mathcal{H} \text{ such that } h' \le h\big\},$$

for a well-defined set of bandwidths $\mathcal{H}$ and a well-chosen bandwidth-dependent sequence $V_n(\cdot)$. In this method, we select a bandwidth by comparing estimators across different scales of bandwidths. For a fixed bandwidth $h \in \mathcal{H}$, the estimator $\widehat{f}_h$ is compared to all estimators $\widehat{f}_{h'}$ with smaller bandwidths $h' \le h$ through their squared L^2-distance. We start the comparison with finer bandwidths having lower bias but higher variance, and the largest bandwidth is selected such that all discrepancies with finer estimators remain below a certain threshold $V_n(h')$. In this way, Lepski's method chooses the smoothest estimator whose bias is still *hidden* by the noise, while the threshold V_n prevents an oversmoothing by detecting when the bias becomes significant.
On the other hand, the novel approach based on [18] named **Penalized Comparison to Overfitting** method (PCO) falls within a similar framework of data-driven pairwise comparison strategies for bandwidth selection. Presented as a simpler method than the one in [19] by *Goldenshluger & Lepski* (GL method) where this time, instead of choosing an optimal bandwidth by a stopping rule as in the classical Lepski's method, we explicitly solve a form of risk-minimization problem. Similarly to a classical Lepski method, the GL procedure again relies on comparing estimators across different bandwidth scales to construct a data-driven proxy for the squared bias

$\sup_{h' \leq h} \left\{ \|\widehat{f}_{h'} - \widehat{f}_h\|_2^2 - V_1(h') \right\}$ (representing a condition on the bias term). The selected bandwidth is then obtained by minimizing the sum of this bias and a variance term $V_2(h)$ in order to achieve the following selection rule

$$\widehat{h} = \underset{h \in \mathcal{H}}{\arg\min} \left\{ \sup_{\substack{h' \in \mathcal{H} \\ h' \leq h}} \left\{ \|\widehat{f}_{h'} - \widehat{f}_h\|_2^2 - V_1(h') \right\}_+ + V_2(h) \right\}. \tag{4.15}$$

This criterion yields an adaptive bias-variance tradeoff and in fact generalizes a classical Lepski method by replacing the stopping rule with an explicit risk-minimization problem. Now a PCO method corresponds to some degenerate case of a GL criterion. Indeed, we start from a classical bias-variance decomposition $\mathbb{E}\|f - \widehat{f}_h\|_2^2 = \|f - f_h\|_2^2 + \mathbb{E}\|f_h - \widehat{f}_h\|_2^2 =: b_h + v_h$ where b_h, v_h are the integrated squared-bias and variance respectively. Then, it turns out that minimizing the risk of $\widehat{f}$ is equivalent to minimizing the criterion $\mathrm{Crit}(h) := \widehat{b}_h + \widehat{v}_h$ for $\widehat{b}_h, \widehat{v}_h$ estimators of b_h, v_h. Now, as we have seen previously (e.g. Proposition (4.2.4)) the integrated variance may be upper-bounded by $\|K\|_2^2/(Nh)$, where they allow to tune this upper-bound with a parameter λ to adjust the contribution of the integrated variance by setting $\widehat{v}_h := \lambda \|K\|_2^2/(Nh)$. Furthermore, denoting by $h_{\min} := \min_{h \in \mathcal{H}} h$ the smallest of the bandwidth grid, *Lacour, Massart & Rivoirard* show that the integrated squared-bias is roughly equal to

$$b_h \approx \|f_{h_{\min}} - f_h\|_2^2 \approx \left\|\widehat{f}_{h_{\min}} - \widehat{f}_h\right\|_2^2 - \frac{\|K_{h_{\min}} - K_h\|_2^2}{N}.$$

As a result, they conclude on a heuristic criterion corresponding to

$$\mathrm{Crit}(h) = \left\|\widehat{f}_{h_{\min}} - \widehat{f}_h\right\|_2^2 + \frac{\lambda \|K_h\|_2^2 - \|K_{h_{\min}} - K_h\|_2^2}{N}. \tag{4.16}$$

Now how does this relate to the GL method? Recall the selection rule (4.15), where under some mild assumptions we can enforce the condition $\sup_{h' \leq h} \|\widehat{f}_{h'} - \widehat{f}_h\|_2^2 \approx \|\widehat{f}_{h_{\min}} - \widehat{f}_h\|_2^2$. Also, supposing that $V_1 = 0$ we now obtain a slightly simpler rule given by

$$\widehat{h} = \underset{h \in \mathcal{H}}{\arg\min} \left\{ \|\widehat{f}_{h_{\min}} - \widehat{f}_h\|_2^2 + \mathrm{pen}_\lambda(h) \right\}. \tag{4.17}$$

As a result, the idea of the authors is relatively simple: instead of comparing estimators $\widehat{f}$ with varying bandwidths $(\widehat{h}, \widehat{h}')$, it is actually sufficient to compare the estimator with a single set of bandwidths $h \in \mathcal{H}$ to the estimator with the smallest one which is set by $h_{\min} := \min_{h \in \mathcal{H}} h$. Mathematically, the method compares the estimators for all $h \in \mathcal{H}$ to the overfitting one $\widehat{f}_{h_{\min}}$ and then adds a penalty term, where $V_2(h) = \mathrm{pen}_\lambda(h)$ is the so-called penalty which is set by the previously defined criterion (4.16) as:

$$\mathrm{pen}_\lambda(h) = \frac{\lambda \|K_h\|_2^2 - \|K_{h_{\min}} - K_h\|_2^2}{N}.$$

We finally mention that the authors show that the case $\lambda = 0$ leads to a minimal penalty, whereas the case $\lambda = 1$ corresponds to an optimal choice of the penalty. In the context of a PCO method for bandwidth selection in a Nadaraya-Watson estimation of the drift function, *Marie & Rosier* [12] use this PCO extension to establish two decision rules for two different bandwidths where one is for the minimization of the risk of $\widehat{bf}_{N,h}$ and the other for the one of $\widehat{f}_{N,h'}$. They are then able to conclude on a global choice of bandwidths $(\widehat{h}, \widehat{h}')$, where the analysis is being made for the adaptive double bandwidths Nadaraya-Watson estimator (see *Comte & Marie* [20])

$$\widehat{b}_{N,\widehat{h},\widehat{h}'}(x) := \frac{\widehat{bf}_{N,\widehat{h}}(x)}{\widehat{f}_{N,\widehat{h}'}(x)} \mathbb{1}_{\left\{\widehat{f}_{N,h'}(x) > \frac{m}{2}\right\}}; \quad x \in [A, B],$$

for which the hope is to independently obtain some good performances for each estimators $\widehat{bf}$ and $\widehat{f}$, and then for the ratio $\widehat{b}$. The overall conclusion of the performance between a LooCV and PCO method is that, in the

nonparametric regression framework the LooCV bandwidth selection method is found to be more satisfactory than the PCO based one. However they still mention that the main advantages of a selection via PCO is that it is easier to implement, is numerically faster and comes with extra theoretical guarantees that a method such as the GL bandwidth selection does not necessarily provide systematically.

4.4 Model Extensions for Space-Time Dependent Diffusions

So far we have only considered time-homogeneous diffusions. A natural question would be to know how we can extend the theory to space-time dependent SDEs. Constructing estimators for these SDEs is considered to be a hard problem since they suffer from a lack of identifiability. Indeed, for time-homogeneous diffusions we know that any statistic computed from the sample paths can be confidently estimated from the available data. However, if we consider space-time dependent SDEs then it is not clear how to separate time variation from space dependence and thus in general we cannot recover $b(t,X_t)$, for instance, from the random curves $\{(t,X_t)\}_t$ (e.g. see [4] for identifiability in SDEs). Let us now mention some extensions on space-time diffusions considered by different authors, where the context and assumptions of the model might differ in some instances.

• A first extension to a space-time setting has been provided by *F.Comte & V.Genon-Catalot* [21] for which they provide a nonparametric drift estimation from N i.i.d. diffusions $(X_t^i)_{1\leq i\leq N}$ given by the inhomogeneous stochastic differential equations

$$dX_t^i = \mu(t,X_t^i)dt + \sigma(t,X_t^i)dW_t^i \quad \text{with } \mu(t,x) := \sum_{k=1}^{K} \alpha_k(t)g_k(x)$$

where $X^i(0)=\eta_i$ for $i\in\{1,\dots,N\}$. This represents a relatively *simple* extension by assuming a space-time diffusion with a linear drift. In this setting, the integer K, the deterministic functions $x\mapsto g_k(x)$ for $k=1,\dots,K$ and $(t,x)\mapsto\sigma(t,x)$ are all known, where the goal is the nonparametric estimation of the K-dimensional functions $t\mapsto\alpha_j(t)\in\mathbb{R}^K$ for $t\geq 0$ and from the continuous observation of the N sample paths through a fixed time interval $[0,T]$. The authors proceed by using a projection method, where each unknown function $\alpha_k(t)$ is estimated by projection onto a finite-dimensional subspace of $L^2([0,T])$.

• Diverging slightly from the context of space-time diffusions, significant attention has been given towards the study of interacting particle systems where sample paths are no longer i.i.d., in particular through McKean-Vlasov models. The diffusions describing these models are not only space-time dependent but also law dependent. These models were for instance studied by *L.Della Maestra & M.Hoffmann* in [22] where they consider the continuous observation of a stochastic system consisting of N interacting particles $(X_t^1,\dots,X_t^N)$ for $t\in[0,T]$ evolving in $\mathbb{R}^d$ such that each particle solves

$$\begin{cases} dX_t^i = b(t,X_t^i,\mu_t^N)dt + \sigma(t,X_t^i)dW_t^i, \;\; 1\leq i\leq N \;\text{ and }\; t\in[0,T] \\ \mathcal{L}(X_0^1,\dots,X_0^N) = \mu_0^{\otimes N}, \end{cases}$$

where $\mu_t^N := N^{-1}\sum_{i=1}^N \delta_{X_t^i}$ is the empirical measure of the system and $\mathcal{L}$ gives the initial law of the process. In this model, $\{W^i\}_i$ are independent Brownian motions and they consider that b,σ are sufficiently regular so that $\mu_t^N\to\mu_t$ in weak convergence such that μ_t is a solution to a parabolic nonlinear equation. The authors then present the construction of a Nadaraya-Watson type estimator on the law dependent drift. The idea is to provide kernel estimators for $\mu_t(x)$ and $b(t,x,\mu_t)$ where a first estimation will be made on the intermediate function $\pi(t,x)=b(t,x,\mu_t)\mu_t(x)$, and thus leads to a quotient estimator to recover b. While an estimator of

$\mu_t(x)$ may be constructed by considering

$$\widehat{\mu}_h^N(t_0,x_0)=\int_{\mathbb{R}^d}K_h(x_0-x)\mu_{t_0}^N(dx)=\frac{1}{N}\sum_{i=1}^N K_h(x_0-X_t^i),$$

such that $(t_0,x_0)\in(0,T]\times\mathbb{R}^d$, $h>0$ with $K_h:=h^{-d}K(h^{-1}x)$, an estimator for the intermediate function requires two kernels over both time and space. Heuristically, we need to define a kernel estimator over the product space $[0,T]\times\mathbb{R}^d$ by considering the random measure $\pi^N(dt,dx):=N^{-1}\sum_{i=1}^N X^i(dt)\delta_{X_t^i}(dx)$ where for a test function $\phi:[0,T]\times\mathbb{R}^d\to\mathbb{R}$ we have

$$\int_{[0,T]\times\mathbb{R}^d}\phi(t,x)\pi^N(dt,dx)=\frac{1}{N}\sum_{i=1}^N\int_0^T\phi(t,X_t^i)dX_t^i.$$

Then for two kernels defined on the product space $(H\otimes K)_{\mathbf{h}}(t,x)=H_{h_1}(t)K_{h_2}(x)$ for $\mathbf{h}=(h_1,h_2),h_i>0$ and $H_h(t)=h^{-1}H(h^{-1}t)$, such that $\int_0^T H(s)ds=\int_{\mathbb{R}^d}K(x)dx=1$, a kernel estimator of $\pi(t_0,x_0)=b(t_0,x_0,\mu_{t_0})\mu_{t_0}(x_0)$ is set to be

$$\widehat{\pi}_{\mathbf{h}}^N(t_0,x_0)=\int_{[0,T]\times\mathbb{R}^d}(H\otimes K)_{\mathbf{h}}(t_0-t,x_0-x)\pi^N(dt,dx)=\frac{1}{N}\sum_{i=1}^N\int_0^T H_{h_1}(t_0-t)K_{h_2}(x_0-X_t^i)dX_t^i.$$

Thus, for some threshold $\omega>0$ preventing the estimator to blow-up for small values of the denominator, a Nadaraya-Watson type estimator for the drift $b(t_0,x_0,\mu_{t_0})$ is given by

$$\widehat{b}_{h,\mathbf{h}}^N(t_0,x_0)=\frac{\widehat{\pi}_{\mathbf{h}}^N(t_0,x_0)}{\widehat{\mu}_h^N(t_0,x_0)\vee\omega},\qquad(t_0,x_0)\in(0,T)\times\mathbb{R}^d.$$

Other authors have studied systems of interacting particles, where for instance *F.Comte & V.Genon-Catalot* [23] study these systems with projection estimators.

• We finally present one last extension, where this time we consider discrete observations of the sample paths instead of continuous observations for the problem of nonparametric estimation of the drift and diffusion coefficients. Such an extension is way more realistic in practice since we never really have access to continuous perfect observations of the sample paths, and so we usually expect to have access to discrete sample points. This novel approach was first presented by *N.Mohammadi, L.V.Santoro & V.M. Panaretos* in [2] where they consider the following general class of time-dependent SDEs

$$dX(t)=(\alpha_0(t)+\mu(t)X(t)^\alpha)dt+\sigma(t)X(t)^\beta dW(t),\tag{4.18}$$

for some numbers α,β. Now as mentioned by the authors, the drift and diffusion coefficients determine the local behaviour of the process, where they can be interpreted as the conditional mean $\mu(t,x)=\lim_{h\to0}h^{-1}\mathbb{E}[X(t+h)-x|X(t)=x]$ and conditional variance $\sigma^2(t,x)=\lim_{h\to0}h^{-1}\mathbb{E}[(X(t+h)-x)^2|X(t)=x]$. Thus if we have access to continuous/dense observations of the sample paths then conducting inference on any class of SDEs can be done directly (for instance, by the work presented in this present paper). However, it is not clear how to adapt this inference procedure to the case where we have discrete/sparse observations and particularly in the presence of noise. The idea presented in [2] is to develop a Functional Data Analysis (FDA) perspective to the problem for which the aim is to perform inference on the law of a continuous time stochastic process, $\{X^i(t)\}_{i=1}^N$, where this sample is taken from the viewpoint of random elements in a separable Hilbert space of functions (usually $L^2([0,T],\mathbb{R})$). As a result, they provide a link between an FDA perspective and the local characteristics of the process by establishing a bijective correspondence between the drift/diffusion coefficients (local perspective) to the mean/covariance functions and their derivatives (global perspective). A starting point to this idea is given as follows: suppose that $\mu(\cdot),\sigma(\cdot)\in C^d([0,1],\mathbb{R})$ for some $d\geq1$ where we denote by $t\mapsto m(t):=\mathbb{E}(X(t))$ the

mean function and $t \mapsto D(t) := \mathbb{E}(X(t)^2)$ the second moment function. Then, since the mean of a stochastic integral yields zero, from (4.18) and the Fubini-Tonelli Theorem we have that

$$\begin{aligned}\mathbb{E}(X(t)) &= \mathbb{E}(X(0)) + \int_0^t (\alpha_0(s) + \mu(s)\mathbb{E}(X(s)^\alpha))ds \\ &\implies \partial_t m(t) = \alpha_0(t) + \mu(t)\mathbb{E}(X(t)^\alpha).\end{aligned}$$

Furthermore, by Itô's Formula and the Fubini-Tonelli Theorem, we derive $X(t)^2$ and its corresponding mean as follows:

$$\begin{aligned}X(t)^2 &= X(0)^2 + \int_0^t \big[2X(s)\big(\alpha_0(s) + \mu(s)X(s)^\alpha\big) + \sigma(s)^2 X(s)^{2\beta}\big]ds + \int_0^t 2\sigma(s)X(s)^{\beta+1}dW(s) \\ &\implies \mathbb{E}(X(t)^2) = \mathbb{E}(X(0)^2) + \int_0^t \big[2\alpha_0(s)\mathbb{E}(X(s)) + 2\mu(s)\mathbb{E}\big(X(s)^{\alpha+1}\big) + \sigma^2(s)\mathbb{E}\big(X(s)^{2\beta}\big)\big]ds \\ &\implies \partial_t D(t) = 2\alpha_0(t)m(t) + 2\mu(t)\mathbb{E}\big(X(t)^{\alpha+1}\big) + \sigma^2(t)\mathbb{E}\big(X(t)^{2\beta}\big).\end{aligned}$$

Hence, with this procedure, the authors were able to relate local properties of the process to a global perspective. In order to obtain some relatively *nice* PDE systems only depending on the mean and second moment function, [2] allow for (4.18) to be defined on $\alpha \in \{0,1\}$, $\beta \in \{0,1/2,1\}$ and for $\alpha_0 \equiv 0$. For example, with $t \in [0,1]$, the PDE system associated to the SDE $dX(t) = \mu(t)X(t)dt + \sigma(t)dW(t)$ is given by

$$\begin{cases}\partial_t m(t) = \mu(t)m(t) \\ \sigma(t)^2 = \partial_t D(t) - 2\mu(t)D(t).\end{cases}$$

Remark 4.4.1. *We should perhaps point out that there are many cases where we can explicitly write a PDE system, however in some instances the estimation procedure goes beyond the scope of the methods presented in [2]. Indeed, considering for example a stochastic Riccati equation (or time-dependent logistic diffusion) $dX(t) = (\mu_1(t)X(t) - \mu_2(t)X(t)^2)dt + \sigma(t)X(t)dW(t)$ it is straightforward to verify that a PDE system is given by*

$$\begin{cases}\partial_t m(t) = \mu_1(t)m(t) - \mu_2(t)D(t) \\ 2\mu_2(t)\mathbb{E}(X(t)^3) = (2\mu_1(t) + \sigma^2(t))D(t) - \partial_t D(t),\end{cases}$$

where it is now unclear how to interpret the estimation of the third moment $\mathbb{E}(X(t)^3)$ using the FDA methods presented in [2].

Of course this first approach is not entirely satisfactory since we are only considering the second order structure $D(t)$ which is restricted to the diagonal segment of the second moment function $\{G(t,t) : 0 \le t \le 1\}$, and so the authors give a more general (averaged) PDE system which depends on the entire second moment function $(s,t) \mapsto G(s,t) := \mathbb{E}(X(s)X(t))$ and its derivatives for all $0 \le s \le t \le 1$, where we omit its precise statement in this essay. The estimation then boils down to estimate the mean and covariance functions in order to obtain plug-in estimators, which is made via a nonparametric local polynomial smoothing method from N i.i.d. diffusion paths.

5 | Conclusion

In this essay, we have studied a Nadaraya-Watson estimator for the nonparametric estimation of the drift function from a SDE diffusion model. This paper started with a brief presentation of local polynomial regression before explicitly constructing the Nadaraya-Watson estimator seen as a solution to a minimisation problem. We have then presented some regularity conditions on the density function $p_t(x_0,x)$ of the SDE via some Kusuoka-Stroock bounds. This led us to establish a Nikol'skii type condition on the *modified* density $f(x) = (T-t_0)^{-1}\int_{t_0}^{T} p_s(x_0,x)ds$, which played a crucial role in the sequel.

These notions laid the technical foundation to construct the nonparametric Nadaraya-Watson estimator for the drift function of a time-homogeneous SDE from the continuous observation of N i.i.d. replications of the process. Using the foundational paper [12] we have derived the corresponding risk-bounds and showed that we could bound the L^2-risk of the estimated drift by a sum (up to constant factors) of the risks of the numerator and denominator from the quotient. We then briefly presented some data-driven methods for bandwidth selection via LooCV and PCO methods. Finally, we discussed some model extensions based on a projection method in [21], an estimator for the drift function for interacting particle systems in [22], and a functional data analysis perspective to the parameter estimation of SDEs given discrete observations of the sample paths in [2].

To conclude, we outline some directions for future work on the topic. In the context of this paper, an interesting extension would be to derive some results in the case of multidimensional diffusion processes although their numerical studies will pose some difficulties. This would entail finding an estimator for a drift $b(t,\mathbf{x})$ with $\mathbf{x} = (x_1,\dots,x_d)$ in a d-dimensional SDE using N i.i.d. sample paths. Furthermore, in the context of discrete/sparse observations of the sample paths, an interesting perspective would be to extend the framework of SDEs in [2] of the form $dX(t) = \mu(t)X(t)^{\alpha}dt + \sigma(t)X(t)^{\beta}dW(t)$, $\alpha \in \{0,1\}$ and $\beta \in \{0,1/2,1\}$, to a larger class of diffusion models. In particular, we could imagine constructing some FDA methods for nonlinear drift and diffusion coefficients, although creating such a global perspective via systems of PDEs is not obviously feasible a priori for a general drift $b(t,x)$ and diffusion $\sigma(t,x)$. Indeed, as we have seen previously in Remark (4.4.1), the PDE system for the nonlinear drift of a stochastic Riccati equation is not solely dependent on the mean function $m(t)$ and second moment function $D(t)$.

Finally, we mention a recent paper published in 2025 by F.Comte & V.Genon-Catalot [24] on some new results for the estimation of space-time dependent drift functions based on a projection method in the Kronecker product notation. This work represents a significant contribution to clarifying the feasibility of estimating space–time dependent drift functions for the continuous observation of the sample paths, and thus paves the way for the study of a broader spectrum of diffusion models.

A | Detailed Proofs

This first appendix is devoted to the details of the proofs of the risk bounds, which were mostly omitted in Section (4.2). While the proofs in this appendix follow *Marie & Rosier* [12], we supplement their arguments with further computational details and expanded explanations. Before presenting these details, we first state two key results that will be used in the proofs.

Lemma A.0.1 (Taylor expansion with integral remainder)**.** *Let $f(x)$ be infinitely differentiable on an interval I around a number α, for $n \geq 0$ and β in the interval I with $\beta \neq \alpha$:*

$$f(\beta) = \sum_{k=0}^{n} \frac{f^{(k)}(\alpha)}{k!}(\beta-\alpha)^k + \int_{\alpha}^{\beta} \frac{f^{(n+1)}(t)}{n!}(\beta-t)^n dt. \tag{A.1}$$

Note that if $n = 0$ this reads $f(\beta) = f(\alpha) + \int_{\alpha}^{\beta} f'(t)dt$, which is just the Fundamental Theorem of Calculus (FTC). This result is actually proven by k integration by parts of the FTC.

Lemma A.0.2 (Generalized Minkowski Inequality)**.** *For any Borel function f on $\mathbb{R}\times\mathbb{R}$ we have:*

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(u,x)du \right)^2 dx \leq \left(\int_{\mathbb{R}} \left(\int_{\mathbb{R}} f^2(u,x)dx \right)^{\frac{1}{2}} du \right)^2. \tag{A.2}$$

A.1 Proof of Proposition (4.2.4)

The idea is to use the decomposition of the L^2-risk into the bias and the variance of $\widehat{f}_{N,h}$. To this end we have:

$$\mathbb{E}\left(\|\widehat{f}_{N,h} - f\|_2^2 \right) = \int_{\mathbb{R}} \mathtt{bias}^2_{\hat{f}}(x)dx + \int_{\mathbb{R}} \mathbb{V}_{\hat{f}}(x)dx.$$

Let us first concentrate on the variance term. Recall that we are considering $X^1, \ldots, X^N$ i.i.d. copies of X and thus,

$$\begin{aligned}
\mathbb{V}_{\hat{f}}(x) &= \mathrm{Var}\left(\frac{1}{N(T-t_0)} \sum_{i=1}^{N} \int_{t_0}^{T} K_h(X_t^i - x)dt \right) \\
&\overset{\text{iid}}{=} \frac{1}{N(T-t_0)^2} \mathrm{Var}\left(\int_{t_0}^{T} K_h(X_t - x)dt \right) \\
&= \frac{1}{N}\mathbb{E}\left[\left(\frac{1}{T-t_0} \int_{t_0}^{T} K_h(X_t - x)dt \right)^2 \right] - \underbrace{\frac{1}{N}\mathbb{E}\left[\frac{1}{T-t_0} \int_{t_0}^{T} K_h(X_t - x)dt \right]^2}_{\geq 0} \\
&\leq \frac{1}{N}\mathbb{E}\left[\left(\frac{1}{T-t_0} \int_{t_0}^{T} K_h(X_t - x)dt \right)^2 \right].
\end{aligned}$$

Now, we use Cauchy-Schwarz to further upper-bound the above expression and use Fubini's Theorem to exchange integration and expectation:

$$\begin{aligned}\mathbb{V}_{\hat{f}}(x) \leq \frac{1}{N}\mathbb{E}\left[\left(\frac{1}{T-t_0}\int_{t_0}^{T} K_h(X_t-x)dt\right)^2\right] &\leq \frac{1}{N(T-t_0)}\int_{t_0}^{T}\mathbb{E}(K_h(X_t-x)^2)dt \\ &= \frac{1}{N(T-t_0)}\int_{t_0}^{T}\int_{\mathbb{R}} K_h(z-x)^2 p_t(x_0,z)dzdt \\ &= \frac{1}{N}\int_{\mathbb{R}} K_h(z-x)^2\left(\frac{1}{T-t_0}\int_{t_0}^{T} p_t(x_0,z)dt\right)dz \\ &= \frac{1}{N}\int_{\mathbb{R}} K_h(z-x)^2 f(z)dz.\end{aligned}$$

By assumption, K is symmetric and thus upon a change of variable $v=(x-z)/h$ we obtain,

$$\begin{aligned}\int_{\mathbb{R}}\mathbb{V}_{\hat{f}}(x)dx \leq \frac{1}{N}\int_{\mathbb{R}}\int_{\mathbb{R}} K_h(z-x)^2 f(z)dzdx &= \frac{1}{Nh^2}\int_{\mathbb{R}}\int_{\mathbb{R}} K\left(\frac{x-z}{h}\right)^2 f(z)dzdx \\ &= \frac{1}{Nh}\underbrace{\left(\int_{\mathbb{R}} f(z)dz\right)}_{=1}\left(\int_{\mathbb{R}} K(v)^2 dv\right) = \frac{\|K\|_2^2}{Nh}.\end{aligned}$$

Let us now find a bound on the integrated squared bias. Recall that $\texttt{bias}_{\hat{f}}(x) = \mathbb{E}\big(\widehat{f}_{N,h}(x)\big) - f(x)$ and also that we are considering $X^1,\dots,X^N$ i.i.d. copies of X. Thus, upon a change of variable $w=(z-x)/h$:

$$\begin{aligned}\texttt{bias}_{\hat{f}}(x) = \mathbb{E}\big(\widehat{f}_{N,h}(x)\big) - f(x) &= \mathbb{E}\left(\frac{1}{N(T-t_0)}\sum_{i=1}^{N}\int_{t_0}^{T} K_h(X_t^i-x)dt\right) - f(x) \\ &\overset{\text{iid}}{=} \frac{1}{T-t_0}\int_{t_0}^{T}\mathbb{E}(K_h(X_t-x))dt - f(x) \\ &= \frac{1}{h(T-t_0)}\int_{t_0}^{T}\int_{\mathbb{R}} K\left(\frac{z-x}{h}\right)p_t(x_0,z)dzdt - f(x) \\ &= \frac{1}{h}\int_{\mathbb{R}} K\left(\frac{z-x}{h}\right)f(z)dz - f(x) \\ &= \int_{\mathbb{R}} K(w)f(wh+x)dw - f(x) = \int_{\mathbb{R}} K(w)\big(f(wh+x)-f(x)\big)dw,\end{aligned}$$

where we have used the fact that the kernel integrates to 1 (i.e. $\int_{\mathbb{R}} K(w)dw = 1$). The idea will now be to use Lemma (3.0.4) on the integrated squared bias. Let us first consider the case $\beta = 1$. Then, by the Generalized Minkowski Inequality (Lemma (A.0.2)) applied with $\theta = hw$ and $k=0$ we find,

$$\begin{aligned}\int_{\mathbb{R}}\texttt{bias}_{\hat{f}}^2(x)dx &= \int_{\mathbb{R}}\left(\int_{\mathbb{R}} K(w)\big(f(wh+x)-f(x)\big)dw\right)^2 dx \\ &\leq \left[\int_{\mathbb{R}} K(w)\left(\int_{\mathbb{R}}\big(f(hw+x)-f(x)\big)^2 dx\right)^{\frac{1}{2}}dw\right]^2 \\ &\leq \left[\int_{\mathbb{R}} K(w)\left(\frac{\Lambda_5}{t_0^{2q(1)}}\big((hw)^2+|hw|^3\big)\right)^{\frac{1}{2}}dw\right]^2 \\ &= h^2\frac{\Lambda_5}{t_0^{2q(1)}}\left[\int_{\mathbb{R}} K(w)|w|(1+|hw|)^{\frac{1}{2}}dw\right]^2 \leq \widehat{\Lambda}_6(t_0)h^2\end{aligned}$$

where $\widehat{\Lambda}_6(t_0)$ is a bounded constant by Assumption (4.2.2) and is given by:

$$\widehat{\Lambda}_6(t_0) = \frac{\Lambda_5}{t_0^{2q(1)}}\left[\int_{\mathbb{R}} |K(w)||w|\big(1+|w|^{\frac{1}{2}}\big)dw\right]^2.$$

Above, we have used the facts that $h \in (0,1]$ and $\sqrt{1+x} \leq 1+\sqrt{x}$ for all $x \geq 0$. Let us now consider the case $\beta \geq 2$. We make use of a Taylor expansion with integral remainder (Lemma (A.0.1)) in the variable (hz) of order $(\beta-2)$ on f (see Remark (A.1.1) for the choice of order), such that for every $w \in \mathbb{R}$,

$$f(hw+x) = f(x) + \underbrace{\mathbb{1}_{\{\beta\geq 3\}}\sum_{k=1}^{\beta-2}\frac{(hw)^k}{k!}f^{(k)}(x)}_{(\dagger)} + \frac{(hw)^{\beta-1}}{(\beta-2)!}\int_0^1 (1-\tau)^{\beta-2}f^{(\beta-1)}(\tau hw+x)d\tau.$$

Now, the idea is to once again apply the Generalized Minkowski Inequality (twice) and use Lemma (3.0.4) with $\theta = hw$ and $k = \beta-1$. Before doing so we remark that since we are considering $v = \beta$, Assumption (4.2.2) is valid for all $k \in \{1,\dots,\beta\}$. In particular, $\int_{\mathbb{R}} w^k K(w)dw = 0$ for all $k \in \{1,\dots,\beta\}$. Thus, since we will be integrating $(\dagger)$ against the kernel $K(w)$, we may freely add the $(\beta-1)$-th term in this sum without changing the result. By noting that $f^{(\beta-1)}(x) = (\beta-1)\int_0^1 (1-\tau)^{\beta-2}f^{(\beta-1)}(x)d\tau$, we then find:

$$\begin{aligned}
\int_{\mathbb{R}} \mathtt{bias}^2_{\hat{f}}(x)dx &= \int_{\mathbb{R}}\left(\int_{\mathbb{R}} K(w)\big(f(wh+x)-f(x)\big)dw\right)^2 dx \\
&= \int_{\mathbb{R}}\left[\int_{\mathbb{R}} K(w)\left(\mathbb{1}_{\{\beta\geq 3\}}\sum_{k=1}^{\beta-1}\frac{(hw)^k}{k!}f^{(k)}(x) + \frac{(hw)^{\beta-1}}{(\beta-2)!}\int_0^1 (1-\tau)^{\beta-2}f^{(\beta-1)}(\tau hw+x)d\tau \right.\right. \\
&\qquad\qquad \left.\left. - \frac{(hw)^{\beta-1}}{(\beta-1)!}f^{(\beta-1)}(x)\right)dw\right]^2 dx \\
&= \int_{\mathbb{R}}\left[\int_{\mathbb{R}} K(w)\left(\frac{(hw)^{\beta-1}}{(\beta-2)!}\int_0^1 (1-\tau)^{\beta-2}\big(f^{(\beta-1)}(\tau hw+x)-f^{(\beta-1)}(x)\big)d\tau\right)dw\right]^2 dx \\
&= \frac{h^{2(\beta-1)}}{|(\beta-2)!|^2}\int_{\mathbb{R}}\left[\int_{\mathbb{R}}\underbrace{K(w)w^{\beta-1}\int_0^1 (1-\tau)^{\beta-2}\big(f^{(\beta-1)}(\tau hw+x)-f^{(\beta-1)}(x)\big)d\tau}_{=:g(x,w)}dw\right]^2 dx.
\end{aligned}$$

Let us now apply the Generalized Minkowski Inequality on $g(x,w)$:

$$\begin{aligned}
\int_{\mathbb{R}} \mathtt{bias}^2_{\hat{f}}(x)dx &\leq \frac{h^{2(\beta-1)}}{|(\beta-2)!|^2}\left[\int_{\mathbb{R}}\left(\int_{\mathbb{R}}\left\{K(w)w^{\beta-1}\int_0^1 (1-\tau)^{\beta-2}\big(f^{(\beta-1)}(\tau hw+x)-f^{(\beta-1)}(x)\big)d\tau\right\}^2 dx\right)^{\frac{1}{2}}dw\right]^2 \\
&= \frac{h^{2(\beta-1)}}{|(\beta-2)!|^2}\left[\int_{\mathbb{R}}|K(w)||w|^{\beta-1}\underbrace{\left(\int_{\mathbb{R}}\left\{\int_0^1 (1-\tau)^{\beta-2}\big(f^{(\beta-1)}(\tau hw+x)-f^{(\beta-1)}(x)\big)d\tau\right\}^2 dx\right)^{\frac{1}{2}}}_{(\Delta)}dw\right]^2.
\end{aligned}$$

We now apply the Generalized Minkowski Inequality a second time on (Δ):

$$(\Delta) \leq \left(\int_0^1 (1-\tau)^{\beta-2}\left\{\int_{\mathbb{R}}\big(f^{(\beta-1)}(\tau hw+x)-f^{(\beta-1)}(x)\big)^2 dx\right\}^{\frac{1}{2}}d\tau\right)^2.$$

Combining these new bounds, we obtain,

$$\int_{\mathbb{R}} \mathtt{bias}^2_{\hat{f}}(x)dx \leq \frac{h^{2(\beta-1)}}{|(\beta-2)!|^2}\left[\int_{\mathbb{R}}|K(w)||w|^{\beta-1}\int_0^1 (1-\tau)^{\beta-2}\left\{\int_{\mathbb{R}}\big(f^{(\beta-1)}(\tau hw+x)-f^{(\beta-1)}(x)\big)^2 dx\right\}^{\frac{1}{2}}d\tau dw\right]^2.$$

We are now in a position where we can apply Lemma (3.0.4) with $\theta = \tau h w$ and $k = \beta - 1$ in order to find,

$$\begin{aligned}\int_{\mathbb{R}} \mathtt{bias}^2_{\hat{f}}(x)dx &\leq \frac{h^{2(\beta-1)}}{|(\beta-2)!|^2}\left[\int_{\mathbb{R}}|K(w)||w|^{\beta-1}\int_0^1(1-\tau)^{\beta-2}\left\{\frac{\Lambda_5}{t_0^{2q(\beta)}}\big((\tau h w)^2+|\tau h w|^3\big)\right\}^{\frac{1}{2}}d\tau dw\right]^2 \\ &= \frac{h^{2\beta}}{|(\beta-2)!|^2}\frac{\Lambda_5}{t_0^{2q(\beta)}}\left[\int_{\mathbb{R}}|K(w)|w|^{\beta}\int_0^1\tau(1-\tau)^{\beta-2}(1+|\tau h w|)^{\frac{1}{2}}d\tau dw\right]^2 \leq \Lambda_6(t_0)h^{2\beta},\end{aligned}$$

such that $\Lambda_6(t_0)$ is a bounded constant by Assumption (4.2.2) and is given by

$$\Lambda_6(t_0) := \frac{\Lambda_5}{|(\beta-2)!|^2 t_0^{2q(\beta)}}\left(\int_{\mathbb{R}}|w|^{\beta}\big(1+|w|^{\frac{1}{2}}\big)|K(w)|dw\right)^2,$$

where we have used the facts that $h \in (0,1]$, $\tau \in [0,1]$ and $\sqrt{1+x} \leq 1+\sqrt{x}$ for all $x \geq 0$. □

Remark A.1.1. *We truncate the Taylor expansion at $k = \beta - 2$ so that the $(\beta-1)$-th derivative of f appears in the remainder term. This is the largest admissible order since Lemma (3.0.4) is only valid for $k \in \{0, ..., \beta-1\}$.*

A.2 Proof of Proposition (4.2.6)

The idea is to use the decomposition of the L^2-risk into the bias and the variance of $\widehat{bf}_{N,h}$. To this end we have:

$$\mathbb{E}\left(\|\widehat{bf}_{N,h} - bf\|_2^2\right) = \int_{\mathbb{R}} \mathtt{bias}^2_{\hat{bf}}(x)dx + \int_{\mathbb{R}} \mathbb{V}_{\hat{bf}}(x)dx.$$

Let us first concentrate on the variance term. Recall that we are considering $X^1, \ldots, X^N$ i.i.d. copies of X and by explicitly substituting the SDE we find,

$$\begin{aligned}\mathbb{V}_{\hat{bf}}(x) &= \mathrm{Var}\left(\frac{1}{N(T-t_0)}\sum_{i=1}^N\int_{t_0}^T K_h(X_t^i - x)dX_t^i\right) \\ &\overset{\text{iid}}{=} \frac{1}{N(T-t_0)^2}\mathrm{Var}\left(\int_{t_0}^T K_h(X_t - x)dX_t\right) \\ &\leq \frac{1}{N}\mathbb{E}\left[\left(\frac{1}{T-t_0}\int_{t_0}^T K_h(X_t - x)dX_t\right)^2\right] \\ &= \frac{1}{N}\mathbb{E}\left[\left(\frac{1}{T-t_0}\int_{t_0}^T K_h(X_t - x)\big(b(X_t)dt + \sigma(X_t)dW_t\big)\right)^2\right] \\ &\leq \frac{2}{N(T-t_0)^2}\mathbb{E}\left[\left(\int_{t_0}^T K_h(X_t - x)b(X_t)dt\right)^2 + \left(\int_{t_0}^T K_h(X_t - x)\sigma(X_t)dW_t\right)^2\right],\end{aligned}$$

where we have used the inequality $(a+b)^2 \leq 2(a^2+b^2)$ in the last inequality. We may apply the Itô Isometry on the second term and also use Cauchy-Schwarz to bound the first term since $K \in L^2$ by assumption. As a result,

$$\begin{aligned}\mathbb{V}_{\hat{bf}}(x) &\leq \frac{2}{N(T-t_0)}\int_{t_0}^T \mathbb{E}\big(K_h(X_t - x)^2 b(X_t)^2\big)dt + \frac{2}{N(T-t_0)^2}\int_{t_0}^T \mathbb{E}\big(K_h(X_t - x)^2\sigma(X_t)^2\big)dt \\ &= \frac{2}{N(T-t_0)}\int_{\mathbb{R}}\int_{t_0}^T K_h(z-x)^2 b(z)^2 p_t(x_0, z)dtdz + \frac{2}{N(T-t_0)^2}\int_{\mathbb{R}}\int_{t_0}^T K_h(z-x)^2\sigma(z)^2 p_t(x_0,z)dtdz\end{aligned}$$

$$= \frac{2}{N}\int_{\mathbb{R}} K_h(z-x)^2 b(z)^2 f(z)dz + \frac{2}{N(T-t_0)}\int_{\mathbb{R}} K_h(z-x)^2\sigma(z)^2 f(z)dz,$$

where we have used Fubini's Theorem to exchange integration and expectation. Moreover, by Assumption (4.2.1) we know that K is symmetric and $b,\sigma \in L^2(\mathbb{R}, f(x)dx)$. Indeed, under Assumptions (4.1.1) and (3.0.2), b,σ have polynomial growth and so for $t\in[0,T]$ we have that $\mathbb{E}(|b(X_t)|^2)$ and $\mathbb{E}(|\sigma(X_t)|^2)$ are bounded. This allows to appropriately bound $\|b\|_{2,f}$ and $\|\sigma\|_{2,f}$ by some finite constant independent of t_0 (see [12, Remark 2.4]). Thus, upon a change of variable $v=(z-x)/h$, the integrated variance is bounded as follows:

$$\begin{aligned}
\int_{\mathbb{R}} \mathbb{V}_{\hat{bf}}(x)dx &\le \frac{2}{N}\int_{\mathbb{R}^2} K_h(z-x)^2 b(z)^2 f(z)dzdx + \frac{2}{N(T-t_0)}\int_{\mathbb{R}^2} K_h(z-x)^2\sigma(z)^2 f(z)dzdx \\
&= \frac{2}{Nh}\int_{\mathbb{R}} b(z)^2 f(z)dz \int_{\mathbb{R}} K(v)^2 dv + \frac{2}{N(T-t_0)h}\int_{\mathbb{R}} \sigma(z)^2 f(z)dz \int_{\mathbb{R}} K(v)^2 dv \\
&= \frac{2\|K\|_2^2}{Nh}\left(\int_{\mathbb{R}} b(z)^2 f(z)dz + \frac{1}{T-t_0}\int_{\mathbb{R}} \sigma(z)^2 f(z)dz\right) \\
&= \frac{2\|K\|_2^2}{Nh}\left(\|b\|_{2,f}^2 + \frac{1}{T-t_0}\|\sigma\|_{2,f}^2\right).
\end{aligned}$$

Let us now find a bound on the integrated squared-bias. As usual, we use the fact that we are considering $X^1,\dots,X^N$ i.i.d. copies of X to find,

$$\begin{aligned}
\texttt{bias}_{\hat{bf}}(x) &= \mathbb{E}\big(\widehat{bf}_{N,h}(x)\big) - b(x)f(x) \\
&= \mathbb{E}\left(\frac{1}{N(T-t_0)}\sum_{i=1}^N \int_{t_0}^T K_h(X_t^i - x)dX_t^i\right) - b(x)f(x) \\
&\overset{\text{iid}}{=} \frac{1}{T-t_0}\mathbb{E}\left(\int_{t_0}^T K_h(X_t-x)dX_t\right) - b(x)f(x) \\
&= \frac{1}{T-t_0}\left[\mathbb{E}\left(\int_{t_0}^T K_h(X_t-x)b(X_t)dt\right) + \underbrace{\mathbb{E}\left(\int_{t_0}^T K_h(X_t-x)\sigma(X_t)dW_t\right)}_{=0}\right] - b(x)f(x) \\
&= \frac{1}{T-t_0}\int_{t_0}^T \mathbb{E}(K_h(X_t-x)b(X_t))dt - b(x)f(x) \\
&= \frac{1}{T-t_0}\int_{t_0}^T \int_{\mathbb{R}} K_h(z-x)b(z)p_t(x_0,z)dzdt - b(x)f(x) \\
&= \int_{\mathbb{R}} K_h(z-x)b(z)f(z)dz - b(x)f(x) = (K_h * (bf))(x) - (bf)(x),
\end{aligned}$$

where in the fourth equality we have used the fact that the Itô integral restricted to $\mathbb{H}^2$ is a martingale-valued map, and thus its expectation yields zero. Therefore, since f is bounded and b belongs to $L^2(\mathbb{R}, f(x)dx)$ we find,

$$\int_{\mathbb{R}} \texttt{bias}_{\hat{bf}}^2(x)dx = \|(K_h * bf) - bf\|_2^2.$$

□

A.3 Proof of Proposition (4.2.7)

The strategy for this proof is to appropriately control the estimator $\widehat{f}_{N,h}$. We thus decompose $\widehat{b}_{N,h}$ in the following way,

$$\widehat{b}_{N,h} - b = \big(\widehat{b}_{N,h} - b\big)\mathbb{1}_{\left\{\widehat{f}_{N,h}(\cdot) > \frac{m}{2}\right\}} + \big(\widehat{b}_{N,h} - b\big)\mathbb{1}_{\left\{\widehat{f}_{N,h}(\cdot) \le \frac{m}{2}\right\}}$$

$$= \left[\frac{\widehat{bf}_{N,h} - bf}{\widehat{f}_{N,h}} + bf\left(\frac{1}{\widehat{f}_{N,h}} - \frac{1}{f}\right)\right] \mathbb{1}_{\left\{\widehat{f}_{N,h}(\cdot) > \frac{m}{2}\right\}} - b\mathbb{1}_{\left\{\widehat{f}_{N,h}(\cdot) \leq \frac{m}{2}\right\}},$$

where we have written $\widehat{b}_{N,h} - b$ over $\left\{\widehat{f}_{N,h}(\cdot) > \frac{m}{2}\right\}$ in a convenient way for the risk-bounds, and used the fact that we force $\widehat{b}_{N,h}$ to be zero over $\left\{\widehat{f}_{N,h}(\cdot) \leq \frac{m}{2}\right\}$. We now consider the norm $\|\cdot\|_{f,A,B} := \|\cdot \mathbb{1}_{[A,B]}\|_{2,f}$ and use the inequality $(a+b)^2 \leq 2(a^2+b^2)$ to find,

$$\left\|\widehat{b}_{N,h} - b\right\|^2_{f,A,B} \leq 2\left\|\left[\frac{\widehat{bf}_{N,h} - bf}{\widehat{f}_{N,h}} + bf\left(\frac{1}{\widehat{f}_{N,h}} - \frac{1}{f}\right)\right] \mathbb{1}_{\left\{\widehat{f}_{N,h}(\cdot) > \frac{m}{2}\right\}}\right\|^2_{f,A,B} + 2\left\|b\mathbb{1}_{\left\{\widehat{f}_{N,h}(\cdot) \leq \frac{m}{2}\right\}}\right\|^2_{f,A,B}. \quad \text{(A.3)}$$

By assumption, for any $x \in [A,B]$, since $f(x) > m$ it is clear that

$$\left|f(x) - \widehat{f}_{N,h}(x)\right| \mathbb{1}_{\left\{\widehat{f}_{N,h}(x) \leq \frac{m}{2}\right\}} \geq \left(f(x) - \widehat{f}_{N,h}(x)\right) \mathbb{1}_{\left\{\widehat{f}_{N,h}(x) \leq \frac{m}{2}\right\}} > m - \frac{m}{2} = \frac{m}{2}. \quad \text{(A.4)}$$

The above inequality will be the basis for the control of the terms involving $\widehat{f}_{N,h}$. Now, let us bound each of the three terms appearing in (A.3).

- For the first term, it is clear that we can bound $\widehat{f}_{N,h}$ by the given condition to find,

$$\left\|\left[\frac{\widehat{bf}_{N,h} - bf}{\widehat{f}_{N,h}}\right] \mathbb{1}_{\left\{\widehat{f}_{N,h}(\cdot) > \frac{m}{2}\right\}}\right\|^2_{f,A,B} \leq \frac{4}{m^2}\left\|\widehat{bf}_{N,h} - bf\right\|^2_{f,A,B} =: \frac{4}{m^2}\left\|\left(\widehat{bf}_{N,h} - bf\right)\mathbb{1}_{[A,B]}\right\|^2_{2,f} \leq \frac{4}{m^2}\left\|\widehat{bf}_{N,h} - bf\right\|^2_{2,f},$$

where this last inequality holds since we do not need to restrict the support of $\widehat{bf}_{N,h} - bf$ to a smaller interval $[A,B]$. We may thus extend the support to any $x \in \mathbb{R}$, where we will then be able to bound the resulting L^2-risk by the result from Proposition (4.2.6).

- We now turn to a bound on the second term by again upper-bounding $\widehat{f}_{N,h}$ with the given condition,

$$\left\|bf\left(\frac{1}{\widehat{f}_{N,h}} - \frac{1}{f}\right) \mathbb{1}_{\left\{\widehat{f}_{N,h}(\cdot) > \frac{m}{2}\right\}}\right\|^2_{f,A,B} = \left\|b\left(\frac{f - \widehat{f}_{N,h}}{\widehat{f}_{N,h}}\right) \mathbb{1}_{\left\{\widehat{f}_{N,h}(\cdot) > \frac{m}{2}\right\}}\right\|^2_{f,A,B} \leq \frac{4}{m^2}\left\|b\left(f - \widehat{f}_{N,h}\right)\right\|^2_{f,A,B}.$$

- We end on a bound of the third term, where an inequality will follow by enlarging the support of the drift b. Indeed, we will use the fact that for two sets S_1 and S_2 we have that $\mathbb{1}_{S_1}(x) \leq \mathbb{1}_{S_1 \cup S_2}(x)$ (this holds since $\mathbb{1}_{S_1 \cup S_2}(x) = \mathbb{1}_{S_1}(x) + \mathbb{1}_{S_2}(x) - \mathbb{1}_{S_1}(x) \cdot \mathbb{1}_{S_2}(x)$ and so the condition $\mathbb{1}_{S_2}(x) - \mathbb{1}_{S_1}(x) \cdot \mathbb{1}_{S_2}(x) \geq 0$ is always satisfied). We thus obtain:

$$\begin{aligned}
\left\|b\mathbb{1}_{\left\{\widehat{f}_{N,h}(\cdot) \leq \frac{m}{2}\right\}}\right\|^2_{f,A,B} &= \left\|b\mathbb{1}_{\left\{f(\cdot) - \widehat{f}_{N,h}(\cdot) \geq f(\cdot) - \frac{m}{2}\right\}}\right\|^2_{f,A,B} \\
&\leq \left\|b\mathbb{1}_{\left\{\left\{f(\cdot) - \widehat{f}_{N,h}(\cdot) \geq f(\cdot) - \frac{m}{2}\right\} \cup \left\{f(\cdot) - \widehat{f}_{N,h}(\cdot) \leq -\left(f(\cdot) - \frac{m}{2}\right)\right\}\right\}}\right\|^2_{f,A,B} \\
&= \left\|b\mathbb{1}_{\left\{\left|f(\cdot) - \widehat{f}_{N,h}(\cdot)\right| \geq f(\cdot) - \frac{m}{2}\right\}}\right\|^2_{f,A,B} \\
&\leq \left\|b\mathbb{1}_{\left\{\left|f(\cdot) - \widehat{f}_{N,h}(\cdot)\right| > \frac{m}{2}\right\}}\right\|^2_{f,A,B},
\end{aligned}$$

where we have used the assumption $f(x) > m$ in the final inequality. Thus, putting everything back together yields,

$$\left\|\widehat{b}_{N,h} - b\right\|^2_{f,A,B} \leq \frac{8}{m^2}\left\|\widehat{bf}_{N,h} - bf\right\|^2_{2,f} + \frac{8}{m^2}\left\|b\left(f - \widehat{f}_{N,h}\right)\right\|^2_{f,A,B} + 2\left\|b\mathbb{1}_{\left\{\left|f(\cdot) - \widehat{f}_{N,h}(\cdot)\right| > \frac{m}{2}\right\}}\right\|^2_{f,A,B}$$

$$:= \frac{8}{m^2}\int_{\mathbb{R}} \big(\widehat{bf}_{N,h} - bf\big)(x)^2 f(x)dx + \frac{8}{m^2}\int_A^B \big(f - \widehat{f}_{N,h}\big)(x)^2 b(x)^2 f(x)dx \\ + 2\int_A^B b(x)^2 f(x)\mathbb{1}_{\left\{\left|f(x)-\widehat{f}_{N,h}(x)\right|>\frac{m}{2}\right\}}dx.$$

Now, by Assumption (3.0.2), we know that f has a sub-Gaussian tail. Also, we have that b is Lipschitz continuous by Assumption (4.1.1) and thus it has also a linear-growth bound of the form $|b(x)| \le C(1+|x|)$. Hence, $|b(x)^2| \le \widetilde{C}(1+|x|^2)$ and so $|(b^2 f)(x)| \le \mathcal{O}\big(|x|^2\exp(-\overline{C}x^2)\big) < \infty$. We thus conclude that $b^2 f$ is bounded on $\mathbb{R}$. As a result, since f and $b^2 f$ are integrable on $\mathbb{R}$ we may further upper-bound as follows,

$$\left\|\widehat{b}_{N,h} - b\right\|^2_{f,A,B} \le \frac{8\|f\|_\infty}{m^2}\left\|\widehat{bf}_{N,h} - bf\right\|_2^2 + \frac{8\|b^2 f\|_\infty}{m^2}\left\|\widehat{f}_{N,h} - f\right\|_2^2 + 2\|b^2 f\|_\infty \int_{\mathbb{R}} \mathbb{1}_{\left\{\left|f(x)-\widehat{f}_{N,h}(x)\right|>\frac{m}{2}\right\}}dx.$$

Finally taking an expectation, using Fubini and Markov's Inequality, we conclude,

$$\begin{aligned}
\mathbb{E}\big(\|\widehat{b}_{N,h} - b\|^2_{f,A,B}\big) &\le \frac{8\|f\|_\infty}{m^2}\mathbb{E}\big(\|\widehat{bf}_{N,h} - bf\|_2^2\big) + \frac{8\|b^2 f\|_\infty}{m^2}\mathbb{E}\big(\|\widehat{f}_{N,h} - f\|_2^2\big) \\
&\qquad + 2\|b^2 f\|_\infty \int_{\mathbb{R}} \mathbb{P}\left(\left|f(x) - \widehat{f}_{N,h}(x)\right| > \frac{m}{2}\right)dx \qquad \rightsquigarrow \text{Fubini} \\
&\le \frac{8\|f\|_\infty}{m^2}\mathbb{E}\big(\|\widehat{bf}_{N,h} - bf\|_2^2\big) + \frac{8\|b^2 f\|_\infty}{m^2}\mathbb{E}\big(\|\widehat{f}_{N,h} - f\|_2^2\big) \\
&\qquad + 2\|b^2 f\|_\infty \int_{\mathbb{R}} \left(\frac{2}{m}\right)^2 \mathbb{E}\big(|\widehat{f}_{N,h}(x) - f(x)|^2\big)dx \qquad \rightsquigarrow \text{Markov} \\
&= \frac{8\|f\|_\infty}{m^2}\mathbb{E}\big(\|\widehat{bf}_{N,h} - bf\|_2^2\big) + \frac{8\|b^2 f\|_\infty}{m^2}\mathbb{E}\big(\|\widehat{f}_{N,h} - f\|_2^2\big) \\
&\qquad + \frac{8\|b^2 f\|_\infty}{m^2}\mathbb{E}\left(\int_{\mathbb{R}} |\widehat{f}_{N,h}(x) - f(x)|^2 dx\right) \qquad \rightsquigarrow \text{Fubini} \\
&= \frac{8\|f\|_\infty}{m^2}\mathbb{E}\big(\|\widehat{bf}_{N,h} - bf\|_2^2\big) + \frac{16\|b^2 f\|_\infty}{m^2}\mathbb{E}\big(\|\widehat{f}_{N,h} - f\|_2^2\big) \\
&\le \frac{8\big(\|f\|_\infty \vee \|b^2 f\|_\infty\big)}{m^2}\big(\mathbb{E}\big(\|\widehat{bf}_{N,h} - bf\|_2^2\big) + 2\mathbb{E}\big(\|\widehat{f}_{N,h} - f\|_2^2\big)\big),
\end{aligned}$$

where upon substitution of the risk-bounds for $\widehat{bf}_{N,h}$ and $\widehat{f}_{N,h}$ from Propositions (4.2.4) and (4.2.6) yields the desired result. □

B | Facts in Nonparametric Regression

B.1 Linearity of Local Polynomial Estimators

Proposition B.1.1. *Suppose that the matrix* $\mathcal{B}_{nx} := \frac{1}{nh}\sum_{i=1}^{n} U\big(\frac{X_i-x}{h}\big)U^T\big(\frac{X_i-x}{h}\big)K\big(\frac{X_i-x}{h}\big) \succ 0$ *is positive definite. Then the local estimator* $\hat{f}_n(x)$ *of* $f(x)$ *is a linear estimator such that* $\hat{f}_n(x) = \sum_{i=1}^{n} Y_i\widetilde{W}_{ni}(x)$ *with* $\widetilde{W}_{ni}(x) := \frac{1}{nh}U^T(0)\mathcal{B}_{nx}^{-1}U\big(\frac{X_i-x}{h}\big)K\big(\frac{X_i-x}{h}\big)$.

Proof. The general steps are from [6] and we complete the details. Consider a fixed x in (2.2) and define the vector $\mathbf{a}_{nx} := \frac{1}{nh}\sum_{i=1}^{n} Y_i U\big(\frac{X_i-x}{h}\big)K\big(\frac{X_i-x}{h}\big)$. As a matter of clarity, we will write $\theta(x) \equiv \theta$. Noting that $\theta^T U\big(\frac{X_i-x}{h}\big) = \theta U^T\big(\frac{X_i-x}{h}\big)$, the key step is to re-write (2.2) as follows:

$$\begin{aligned}
\hat{\theta}_n(x) &= \underset{\theta\in\mathbb{R}^{k+1}}{\arg\min}\sum_{i=1}^{n}\left[Y_i - \theta^T U\left(\frac{X_i-x}{h}\right)\right]^2 K\left(\frac{X_i-x}{h}\right)\\
&= \underset{\theta\in\mathbb{R}^{k+1}}{\arg\min}\sum_{i=1}^{n}\left[Y_i^2 - 2Y_i\theta^T U\left(\frac{X_i-x}{h}\right) + \left(\theta^T U\left(\frac{X_i-x}{h}\right)\right)^2\right] K\left(\frac{X_i-x}{h}\right)\\
&= \underset{\theta\in\mathbb{R}^{k+1}}{\arg\min}\left[-2\theta^T\sum_{i=1}^{n} Y_i U\left(\frac{X_i-x}{h}\right)K\left(\frac{X_i-x}{h}\right) + \theta^T\left(\sum_{i=1}^{n} U\left(\frac{X_i-x}{h}\right)U^T\left(\frac{X_i-x}{h}\right)K\left(\frac{X_i-x}{h}\right)\right)\theta\right]\\
&= \underset{\theta\in\mathbb{R}^{k+1}}{\arg\min}\Big(-2nh\theta^T\mathbf{a}_{nx} + nh\theta^T\mathcal{B}_{nx}\theta\Big) = \underset{\theta\in\mathbb{R}^{k+1}}{\arg\min}\Big(-2\theta^T\mathbf{a}_{nx} + \theta^T\mathcal{B}_{nx}\theta\Big).
\end{aligned}$$

Note that, in the third equality, we have used the fact that the term $\sum_{i=1}^{n} Y_i^2 K\big(\frac{X_i-x}{h}\big)$ is independent of θ, and thus will not influence the minimisation over θ. Let us now explicitly find a necessary condition for this minimisation. To this end, suppose that we have found the optimal $\hat{\theta}$, and as a result we must have that $\nabla_{\hat{\theta}}\big(-2\hat{\theta}^T\mathbf{a}_{nx} + \hat{\theta}^T\mathcal{B}_{nx}\hat{\theta}\big) = 0$. Let us now expand this gradient:

$$\nabla_{\hat{\theta}}\Big(-2\hat{\theta}^T\mathbf{a}_{nx} + \hat{\theta}^T\mathcal{B}_{nx}\hat{\theta}\Big) = -2\nabla_{\hat{\theta}}\Big(\hat{\theta}^T\mathbf{a}_{nx}\Big) + \nabla_{\hat{\theta}}\Big(\hat{\theta}^T\mathcal{B}_{nx}\hat{\theta}\Big) = -2\mathbf{a}_{nx} + \Big(\mathcal{B}_{nx} + \mathcal{B}_{nx}^T\Big)\hat{\theta} = 2\big(\mathcal{B}_{nx}\hat{\theta} - \mathbf{a}_{nx}\big) = 0,$$

where we have used the fact that $\mathcal{B}_{nx}$ is symmetric, i.e. $\mathcal{B}_{nx} = \mathcal{B}_{nx}^T$. This now gives the necessary condition $\mathcal{B}_{nx}\hat{\theta}_n(x) = \mathbf{a}_{nx}$. Now since we assume that the matrix $\mathcal{B}_{nx}$ is positive definite, we know that it has an inverse and as a result the $LP(k)$ estimator is unique and is given by $\hat{\theta}_n(x) = \mathcal{B}_{nx}^{-1}\mathbf{a}_{nx}$ (a necessary and sufficient condition for the minimisation). Finally, by a direct application of (2.3), we conclude that:

$$\begin{aligned}
\hat{f}_n(x) = U^T(0)\hat{\theta}_n(x) &= U^T(0)\mathcal{B}_{nx}^{-1}\left(\frac{1}{nh}\sum_{i=1}^{n} Y_i U\left(\frac{X_i-x}{h}\right)K\left(\frac{X_i-x}{h}\right)\right)\\
&= \sum_{i=1}^{n} Y_i\left(\frac{1}{nh}U^T(0)\mathcal{B}_{nx}^{-1}U\left(\frac{X_i-x}{h}\right)K\left(\frac{X_i-x}{h}\right)\right) \equiv \sum_{i=1}^{n} Y_i\widetilde{W}_{ni}(x).
\end{aligned}$$

□

B.2 NW Estimator - Solution of a Minimisation Problem

Proposition B.2.1. *Let $K : \mathbb{R} \to \mathbb{R}$ be a kernel and $h > 0$ be a bandwidth. The Nadaraya-Watson estimator is given by:*

$$f_n^{NW}(x) = \frac{\sum_{i=1}^n Y_i K\left(\frac{X_i - x}{h}\right)}{\sum_{i=1}^n K\left(\frac{X_i - x}{h}\right)} \cdot \mathbb{1}_{\left\{\sum_{i=1}^n K\left(\frac{X_i - x}{h}\right) \neq 0\right\}}. \tag{B.1}$$

Proof. We explicitly solve the minimisation problem, and find:

$$\begin{aligned} f_n^{NW}(x) = \operatorname*{arg\,min}_{\theta \in \mathbb{R}} \sum_{i=1}^n (Y_i - \theta)^2 K\left(\frac{X_i - x}{h}\right) &= \operatorname*{arg\,min}_{\theta \in \mathbb{R}} \sum_{i=1}^n \left(Y_i^2 - 2\theta Y_i + \theta^2\right) K\left(\frac{X_i - x}{h}\right) \\ &= \operatorname*{arg\,min}_{\theta \in \mathbb{R}} \left(-2\theta \sum_{i=1}^n Y_i K\left(\frac{X_i - x}{h}\right) + \theta^2 \sum_{i=1}^n K\left(\frac{X_i - x}{h}\right)\right). \end{aligned}$$

For the optimal $\hat{\theta}$, it must be that

$$\frac{\partial}{\partial \hat{\theta}} f_n^{NW}(x) = -2 \sum_{i=1}^n Y_i K\left(\frac{X_i - x}{h}\right) + 2\hat{\theta} \sum_{i=1}^n K\left(\frac{X_i - x}{h}\right) = 0.$$

Solving this equation for $\hat{\theta}$ yields the result. Finally, it is clear that $\partial_{\hat{\theta}}^2 f_n^{NW}(x) = 2\sum_{i=1}^n K\left(\frac{X_i - x}{h}\right) \geq 0$ and so we have a minimum. □

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